Two Hidden Layers Suffice: The Caterpillar Net

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Abstract

Every continuous piecewise linear (CPWL) function is a finite signed sum of maxima of affine functions, making maximum representation a central problem in the study of the expressiveness of ReLU networks. Earlier work disproved the conjectured base-two depth bound, leaving open whether a constant number of hidden layers suffices to represent the maximum. Brunck, Hertrich, Rueß, Stade, and Yehudayoff first established that two hidden layers suffice for all CPWL functions. Our smaller caterpillar construction was developed independently and concurrently with their subsequent combinatorial construction. We give the *Caterpillar Net*, a non-recursive explicit construction representing the maximum of N real inputs exactly with two hidden layers of widths $\binom{N}{2} + N + 1$ and $2^{O(N \log N)}$, rational weights, and zero biases. Every expansion of the maximum in its finite vocabulary compiles with the same first hidden layer. Sharing its first hidden layer across a Wang–Sun representation gives two-hidden-layer networks for all CPWL functions. [1]

1 Introduction

In this paper we investigate the representation capabilities of ReLU networks: what functions are representable by such networks of different depths and widths? ReLU networks represent precisely the continuous piecewise-linear (CPWL) functions with finitely many pieces [ABMM18]. A sharp question is therefore: given a CPWL function, what is the smallest depth, or the smallest width at a prescribed depth, needed to represent it exactly [HBD+23, Saf26]? The maximum is a universal building block for such functions: every CPWL function is a finite signed sum of maxima of affine functions [WS05]. Write

$$\max_N(x) := \max_{0 \le i < N} x_i. \tag{1}$$

Theorem (Wang–Sun, [WS05], `Lean_1`)**.** *Let $d \ge 1$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous and affine on each member of a finite polyhedral cover. There are $R \in \mathbb{N}$, affine maps and signs*

$$A_t : \mathbb{R}^d \longrightarrow \mathbb{R}^{d+1}, \qquad \varepsilon_t \in \{-1, 1\} \qquad (1 \le t \le R) \tag{2}$$

such that

$$f(x) = \sum_{t=1}^{R} \varepsilon_t \max_{d+1}(A_t(x)) \qquad \textit{for every } x \in \mathbb{R}^d. \tag{3}$$

[1] Our results were obtained over months of proof search in Lean using Claude Code and Codex; the AI methodology statement describes their roles. The Lean code is available in the accompanying repository https://github.com/JayPiZimmermann/Max_Caterpillar. Theorem titles link to the Lean formalization; clickable equation numbers link to supporting declarations or written proofs.

Thus a two-hidden-layer representation of $\max_{d+1}$ gives one for every CPWL function on $\mathbb{R}^d$: precompose with the affine maps and add the networks in parallel. Section 7 reuses their first hidden layer; the representation theorems and the resulting networks are formalized.

Our contribution. The *Caterpillar Net* gives a nonrecursive two-hidden-layer representation of the maximum with explicit bounds on both hidden-layer widths. The construction uses a finite dictionary of support functions of caterpillar polytopes and their faces. The maximum is the support function of a simplex. We first express this support function as an alternating sum of caterpillar support functions, up to a linear combination of support functions of lower-dimensional polytopes. We then represent the remaining error using faces of the same caterpillar polytopes and elementary ReLU features. Normalizing and merging these descriptions yields a width bound for the complete network, including all correction terms.

Write $\sigma(t) := \max(0, t)$ applied coordinatewise to vectors. A network has an input layer of N coordinates, first and second hidden layers of widths h_1 and h_2, respectively, and a scalar output layer. With real weight matrices W_1, W_2, output vector w_3, and biases b_1, b_2, b_3 of the corresponding dimensions, it evaluates as

$$x \longmapsto w_3^\top \sigma\big(W_2\sigma(W_1x + b_1) + b_2\big) + b_3. \tag{4}$$

There are no skip connections; equality is required on all of $\mathbb{R}^N$. A network has *rational weights* if every entry of W_1, W_2, and w_3 is rational, and *zero biases* if b_1, b_2, and b_3 vanish.

Theorem 1 (Caterpillar Net, `Lean_2`, Proof)**.** *For every $N \geq 3$, there is a two-hidden-layer ReLU network with rational weights and zero biases computing $\max_N$ with widths*

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = 3^{N-2}C_N - N(N-1) + 2, \tag{5}$$

where

$$C_N := \sum_{m=2}^{N-1} \binom{N}{m} \frac{m!}{2}(N-m)m^{N-m-1}. \tag{6}$$

Thus $h_1 = O(N^2)$ and $h_2 = 2^{O(N \log N)}$ as $N \to \infty$. The rational coefficients of this construction are specified by the finite linear system in Equation (24)*.*

Further results. The construction chooses one basis of the caterpillar dictionary extending a basis of its auxiliary span, determines its coefficients by rational evaluations, and compiles them into the network in Section 3.2. The basis has q_N nonauxiliary functions, so this same network has $q_N + 2$ neurons in its second hidden layer before zero padding to the stated width. Conversely, Section 4 gives exponential lower bounds for caterpillar expansions and proves $q_N \geq (N-2)!$. Section 7 applies the construction to sums of maxima.

2 Related work

The conjecture that $\max\{0, x_1, \ldots, x_{2^k}\}$ requires more than k hidden layers [HBD+23] was disproved for $k = 2$ by the maximum of five inputs in [BBH+26]. Exact computer-assisted searches subsequently obtained two-hidden-layer representations up to $N = 12$ in [RAB+26] and, independently, up to $N = 8$ in [WB26]. The improved general depth bounds remained logarithmic, leaving constant depth open.

Setting all biases to zero does not change a network that computes a positively homogeneous function such as $\max_N$ [HBD+23, Proposition 2.3], [HHL23, Lemma 7], so biases give no additional power for

representing the maximum. Depth lower bounds hold for integer weights [HHL23] and rational weights whose denominators are restricted to powers of a fixed integer [AHM25]. Our weights are rational, but their denominators are not restricted in this way, so neither bound applies. Other results concern braid-fan-compatible networks [GHL25] and first hidden layer width bounds [Saf26, Lu26].

Theorem 2 (Brunck, Hertrich, Rueß, Stade, and Yehudayoff, [BHR+26])**.** *Every continuous piecewise affine function $f : \mathbb{R}^d \to \mathbb{R}$ with finitely many affine pieces can be represented exactly by a ReLU network with two hidden layers.*

Their first proof proceeds by induction on dimension, using homological algebra to construct a linear combination of support functions of pyramids over zonotopes that agrees with the simplex support function modulo lower-dimensional terms. Our construction restricts these building blocks to an explicitly specified finite family, the caterpillar polytopes; we deliver a geometric proof of the corresponding identity.

Independently and concurrently, they developed a second proof based on combinatorial cancellation, with building blocks closely related to ours. Appendix D identifies them exactly: for nonempty marked sets their pyramids are translated caterpillar polytopes, and the remaining pyramids are rooted stars, which are not caterpillar functions. The comparison identifies their building blocks and faces; it does not identify their final correction coefficients. We establish explicit width bounds for the first and second hidden layers; their paper states no widths.

3 The Caterpillar Net

Two identities drive the construction:

$$\max(s, t) = t + \sigma(s - t), \tag{7}$$

$$\sigma(\ell(x)) = \sigma(-\ell(x)) + \ell(x). \tag{8}$$

The first suggests a path with attached leaves. A signed sum of the resulting caterpillar support functions agrees with the maximum modulo lower-dimensional terms; proper faces and linear functions correct the error. The second identity reverses hinges to merge units in Section 6.

Fix $N \geq 3$ and write $x = (x_0, \ldots, x_{N-1}) \in \mathbb{R}^N$.

Definition 3 (Graph, forest, and tree functions)**.** For distinct labels i, j, put

$$\ell_{ij}(x) := x_i - x_j. \tag{9}$$

Let L be a set of these comparisons, containing at most one orientation of each unordered pair. Given a real linear head $h(x) = \sum_i h_i x_i$ and real coefficients $c = (c_\ell)_{\ell \in L}$, define the *graph function*

$$U_{h,L,c}(x) := \sigma\left(h(x) + \sum_{\ell \in L} c_\ell \sigma(\ell(x)) \right). \tag{10}$$

Its *comparison graph* G_L has all N input labels as vertices and an edge $\{i, j\}$ for each $\ell_{ij} \in L$; the head is not an edge. We call this a *forest function* when G_L is acyclic, and a *tree function* when its nonisolated part is a tree. A *pseudoforest function* allows at most one cycle in each component. Isolated labels remain in all induced-graph and component counts. The graph is part of the description: zero coefficients may be retained in L when a bound concerns the original graph. Opposite orientations of the same comparison can be combined using Equation (8).

Write $U_{h,L}$ when every $c_\ell = 1$, and call (h, L) its *head–hinge pair*. For such a unit-weight forest function, set $\kappa_i = h_i + |\{j : \ell_{ij} \in L\}|$. A degree-one vertex with $\kappa_i = 0$ is a *pendant*. The *reduced forest* is obtained by deleting all pendants simultaneously; newly created leaves are not deleted again.

Definition 4 (Canonical trees and face functions)**.** A *canonical tree function* is $U_{\ell_{ra},L}$, where a is the apex label, $r \neq a$ is the head root, and L is an arbitrary orientation of a tree on all labels other than a. If every edge of an underlying tree T points away from a source s, write $U_{a,s,r,T}$ and call the function *source-oriented*. Use the same indices on its face functions, $U^{\tau}_{a,s,r,T}$. A *rooted-star function* has a star centred at its head root, with all edges directed away from that root. Write

$$S_{a,c}(x) := \sigma\Big(x_c - x_a + \sum_{l \notin \{a,c\}} \sigma(x_c - x_l)\Big). \tag{11}$$

An *oriented-star function* allows any edge orientations and any head root in the star.

For a graph function, an *edge state* $\tau(\ell) \in \{0, \ell, \sigma \circ \ell\}$ omits an edge, keeps it linearly, or retains $x \mapsto \sigma(x_i - x_j)$, respectively. Put

$$U^{\tau}_{h,L,c}(x) := \sigma\Big(h(x) + \sum_{\ell \in L} c_\ell \tau(\ell)(x)\Big). \tag{12}$$

Write $U^{\tau}_{h,L}$ for unit coefficients. For canonical trees these are the *face functions*, with the geometric interpretation below. Every edge-state function is already a graph function from Definition 3:

$$U^{\tau}_{h,L,c} = U_{h',M,c|_M}, \qquad h' := h + \sum_{\tau(\ell)=\ell} c_\ell \ell, \qquad M := \{\ell \in L : \tau(\ell) = \sigma \circ \ell\}. \tag{13}$$

Even when the original comparisons form a tree, the retained graph G_M can be disconnected: omitting or linearizing a tree edge splits its component. The linear state also changes the head, as in Equation (13). Thus tree face functions are forest functions, with heads determined by their original tree and edge states. For unit weights their reduced forests are computed from (h', M). Equivalently, one may keep L and give every nonretained comparison coefficient zero; this describes the same function with the original graph available for bounds on its matching number.

Definition 5 (Labelled caterpillar, `Lean_3`)**.** A *caterpillar description* is a permutation–set pair

$$\pi \in \mathrm{Sym}\{0, \dots, N-1\}, \qquad \varnothing \neq S \subseteq \{1, \dots, r\}, \qquad r := N - 2. \tag{14}$$

Position j carries label $\pi(j+1)$, and $\pi(0)$ is the *apex*. The *spine positions* are $\{0\} \cup S$. The *cut set* S separates positions $j-1$ and j whenever $j \in S$. The last spine position q and the *parent* $p(j)$, the nearest preceding spine position, are

$$q := \max S, \qquad p(j) := \max\big(\{0\} \cup \{i \in S : i < j\}\big) \qquad (1 \le j \le r). \tag{15}$$

The parent–child edges form the *labelled caterpillar*: a path through the spine positions with each remaining label attached as a leaf to its parent (Figure 1). The apex is separate from the tree.

Definition 6 (Caterpillar functions, `Lean_4`, `Lean_5`)**.** For a caterpillar description (π, S), use the comparisons and head

$$L_{\pi,S} := \big\{\ell_{\pi(p(j)+1),\pi(j+1)} : 1 \le j \le r\big\}, \tag{16}$$

$$h_{\pi,S} := \ell_{\pi(q+1),\pi(0)}. \tag{17}$$

These give a canonical tree function with apex $\pi(0)$, source $\pi(1)$, and head root $\pi(q+1)$. Its *caterpillar function* and *caterpillar face functions* are the specializations

$$U_{\pi,S} := U_{h_{\pi,S},L_{\pi,S}}, \qquad U^{\tau}_{\pi,S} := U^{\tau}_{h_{\pi,S},L_{\pi,S}}. \tag{18}$$

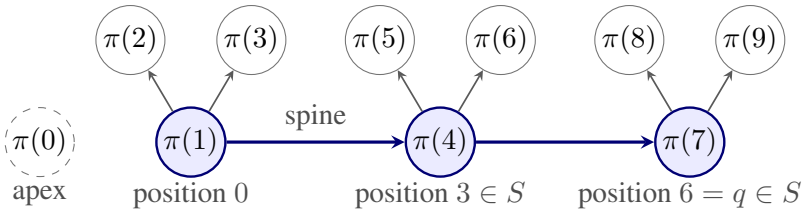


Figure 1: A labelled caterpillar with $N = 10$ and $S = \{3, 6\}$. The spine $\pi(1) \to \pi(4) \to \pi(7)$ has two leaves at each vertex; all arrows point from parent to child. The apex $\pi(0)$ is separate from the tree. The last spine position is $q = 6$, so the head function is $h_{\pi,S}(x) = x_{\pi(7)} - x_{\pi(0)}$.

The term *face function* comes from the following geometry. Identify $a(x) = \langle a, x\rangle$ with its coefficient vector. Then $\ell_{ij} = e_i - e_j$ points from e_j to e_i, and $h = h_{\pi,S} = e_{\pi(q+1)} - e_{\pi(0)}$ points from the apex label to the last spine label. Each $\ell \in L = L_{\pi,S}$ points from child to parent. As free vectors, h places a base corner and the ℓ generate its edges. The polytope apex is 0, distinct from the labelled point $e_{\pi(0)}$. Define the *support function* and the caterpillar polytope by

$$h_P(x) := \max_{p\in P}\langle p, x\rangle, \tag{19}$$

$$Q_{h,L} := \operatorname{conv}\Big(\{0\} \cup \Big(h + \sum_{\ell\in L}[0, \ell]\Big)\Big), \qquad U_{h,L} = h_{Q_{h,L}}. \tag{20}$$

Indeed, the segment $[0, \ell]$ has support function $\sigma \circ \ell$, Minkowski sums add support functions, and adjoining 0 applies the outer ReLU. The base is the translated zonotope $h + \sum_{\ell\in L}[0, \ell]$. For a state τ in Equation (12), set $t_\ell = \mathbb{1}_{\{\tau(\ell)=\ell\}}$, where $\mathbb{1}$ denotes the indicator. It is zero for omitted edges and hinges; the latter contribute segments. The corresponding polytope is $Q^\tau_{h,L} := \operatorname{conv}\big(\{0\} \cup (h + \sum_{\ell\in L} t_\ell \ell + \sum_{\tau(\ell)=\sigma\circ\ell}[0, \ell])\big)$, and $U^\tau_{h,L} = h_{Q^\tau_{h,L}}$. Figure 3 illustrates the translated generators for $N = 4$ on the slice $x_0 = 0$, where a becomes (a_1, a_2, a_3) and e_0 is represented by 0.

For a canonical tree, the head connects the apex label to the tree, so the head and $N - 2$ edge functions are linearly independent. Thus the base is an $(N - 2)$-dimensional zonotope with affine hull avoiding 0, and $Q_{h,L}$ is an $(N - 1)$-dimensional pyramid. Each nontrivial face containing 0 cones over a base face. Maximizing on each base segment selects 0, ℓ, or $[0, \ell]$, with support functions 0, ℓ, and $\sigma \circ \ell$; the outer ReLU adjoins the apex. These are precisely the three states in Equation (12).

Definition 7 (Dictionary, `Lean_6`)**.** The *dictionary* $\mathcal{D}$ is the finite set of functions $\mathbb{R}^N \to \mathbb{R}$ consisting of every face function $U^\tau_{\pi,S}$, over all caterpillar descriptions (π, S) and all states τ, together with the *auxiliary functions*

$$\begin{aligned} &x \mapsto \sigma(x_i), \quad x \mapsto \sigma(-x_i) \quad (0 \le i < N),\\ &x \mapsto \sigma(x_i - x_j) \qquad\qquad (0 \le i < j < N). \end{aligned} \tag{21}$$

Caterpillar heads and edge functions are coordinate differences, so these face functions and the functions $x \mapsto \sigma(x_i - x_j)$ are translation invariant: with $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^N$,

$$U^\tau_{\pi,S}(x + t\mathbf{1}) = U^\tau_{\pi,S}(x) \qquad (t \in \mathbb{R}). \tag{22}$$

Only $\sigma(\pm x_i)$ change under this translation. Theorem 16 proves that $\mathcal{D}$ spans the maximum.

3.1 Rational coefficients

Let $V_N = \operatorname{span} \mathcal{D}$ and let A_N be the span of the auxiliary functions. Choose a basis $g_1, \dots, g_a$ of A_N from those functions and extend it by caterpillar face functions to a basis $g_1, \dots, g_d$ of V_N. Put

$$q_N = \dim(V_N/A_N) = d - a. \tag{23}$$

Choose rational inputs $y_1, \dots, y_d$ for which the evaluation matrix $E = (g_j(y_i))_{i,j}$ is invertible, and define

$$c = E^{-1}\big(\max_N(y_1), \dots, \max_N(y_d)\big)^\top. \tag{24}$$

These are the coefficients used by the Caterpillar Net.

Lemma 8 (The coefficient identity)**.** *The rational inputs can be chosen as stated, $c \in \mathbb{Q}^d$, and*

$$\max_N(x) = \sum_{j=1}^{d} c_j g_j(x) \qquad (x \in \mathbb{R}^N). \tag{25}$$

Proof. The vectors $(g_1(y), \dots, g_d(y))$, $y \in \mathbb{Q}^N$, span $\mathbb{R}^d$: otherwise a nonzero linear functional annihilating them would give a nontrivial combination of the g_j vanishing on the dense set $\mathbb{Q}^N$, hence everywhere by continuity. This contradicts independence. Select d independent evaluation vectors as the rows of E. All entries of E and the right side of Equation (24) are rational, so c is rational. By Theorem 16, $\max_N \in V_N$; its unique basis coordinates solve this same invertible system and therefore equal c. □

The first a terms form the auxiliary remainder R; the other q_N terms give the retained face functions. We retain all these basis functions, including any with zero coefficient. Distinct basis functions cannot normalize to the same description, so $|K| = q_N$ below. The following normalization and network assembly use these fixed rational coefficients.

3.2 The network

The network uses the rational coefficients selected above. Equation (13) writes each selected face function as $U_{h',M}$. By Equation (8), reversing each remaining ℓ_{ij} with $i > j$ gives the same function with the *normalized description*

$$\begin{aligned} M^+ &:= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in M\}, \\ h^+ &:= h' + \sum_{\ell_{ij} \in M,\ i>j} \ell_{ij}. \end{aligned} \tag{26}$$

Geometrically, $[0, \ell] = \ell + [0, -\ell]$: reversing a generator shifts the head by the old generator. Normalization chooses a consistent orientation and a corresponding base corner; it does not rescale vectors or change the polytope (Figure 3). Write $U_k := U_{h,M}$ for a normalized description $k = (h, M)$, and let $\mathcal{K}_0$ be the set of normalized descriptions of all caterpillar face functions.

Definition 9 (Expansions)**.** An *expansion* of $\max_N$ is an identity $\max_N = \sum_{k \in K} d_k U_k + R$ with finitely many graph functions U_k, real coefficients d_k, and $R \in A_N$, unless a larger remainder space is specified. Its size is $|K|$. When $K \subseteq \mathcal{K}_0$ indexes normalized caterpillar face functions, it is a *caterpillar expansion* (`Lean_7`). A *shallow* remainder is a linear combination of linear functions and functions $x \mapsto \sigma(\ell(x))$ with ℓ linear.

The network has three parts (Figure 2).

- The *first hidden layer* computes the shared features

$$\Phi(x) := \big(\sigma(x_0),\ \sigma(-x_0),\ (\sigma(x_i - x_0))_{i>0},\ (\sigma(x_i - x_j))_{i<j}\big). \tag{27}$$

- The *second hidden layer* computes the retained units U_k, $k \in K$, for a set $K \subseteq \mathcal{K}_0$, and the two neurons $\sigma(R)$ and $\sigma(-R)$ for an auxiliary remainder $R \in A_N$.
- The *output layer* returns $\sum_{k\in K} d_k U_k + \sigma(R) - \sigma(-R)$.

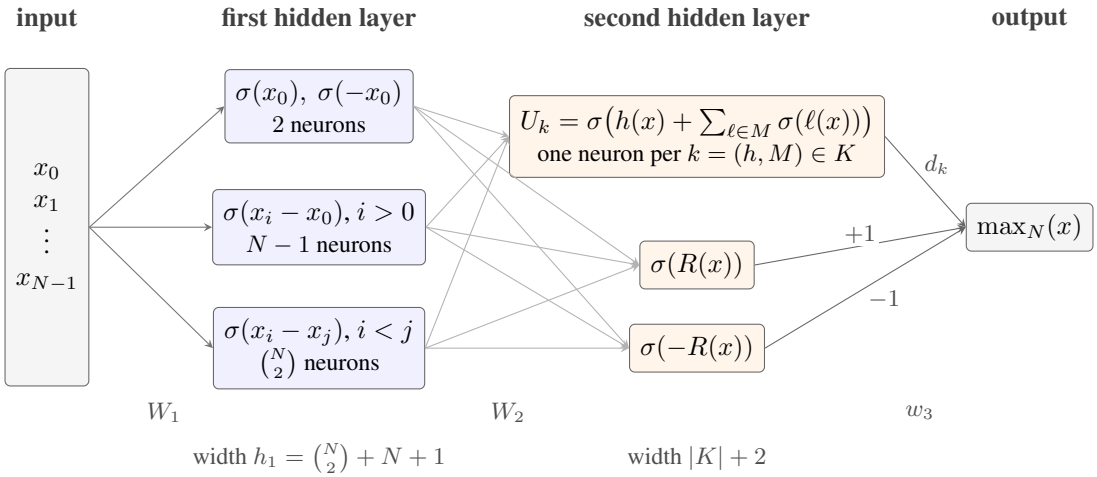


Figure 2: The Caterpillar Net. The first hidden layer is fixed. The second hidden layer has one neuron for every retained normalized description $k = (h, M)$ and two neurons for the auxiliary remainder R. All biases are zero; the matrices are given in Equation (34).

Lemma 10 (Network assembly, `Lean_8`, Proof)**.** *Normalize the q_N chosen face functions and collect their descriptions in $K \subseteq \mathcal{K}_0$. Let d_k be their coefficients from Equation* (24)*, and let R contain its auxiliary terms. Then*

$$\max_N = \sum_{k\in K} d_k U_k + R. \tag{28}$$

The network with first hidden layer Equation (27) *and the matrices in Equation* (34) *computes* $\max_N$*. It has widths h_1 and $|K| + 2 = q_N + 2$, rational weights, and zero biases.*

Proof of Lemma 10. The first hidden layer outputs Φ from Equation (27). Thus W_1 has rows $e_0^\top$ and $-e_0^\top$, then $e_i^\top - e_0^\top$ for $i > 0$, then $e_i^\top - e_j^\top$ for $i < j$, using a fixed ordering within each block. Its row count is

$$\binom{N}{2} + (N-1) + 2 = \binom{N}{2} + N + 1 = h_1. \tag{29}$$

By Equation (8), these features recover all coordinates,

$$\begin{aligned} x_0 &= \sigma(x_0) - \sigma(-x_0), \\ x_i &= x_0 + \sigma(x_i - x_0) - \sigma(x_0 - x_i) \qquad (i > 0). \end{aligned} \tag{30}$$

For each $k = (h, M) \in K$, substitute Equation (30) into $h(x) + \sum_{\ell \in M} \sigma(\ell(x))$ and collect the coefficients of $\Phi(x)$ into a row vector $w_k^\top$. Every hinge in M already has the orientation $i < j$, so its coefficient is added directly to the corresponding feature.

To specify the remaining row, collect the auxiliary coefficients of R as

$$R(x) = \sum_i p_i \sigma(x_i) + \sum_i n_i \sigma(-x_i) + \sum_{i<j} s_{ij} \sigma(x_i - x_j). \tag{31}$$

Each retained unit is translation invariant by Equation (22), and $\max_N(x + t\mathbf{1}) = \max_N(x) + t$, so Equation (28) gives

$$R(x + t\mathbf{1}) = R(x) + t, \qquad \sum_i p_i = 1. \tag{32}$$

The coefficient sum follows by setting $x = 0$ and $t = 1$. Now choose t large enough that every $x_i + t$ is nonnegative. Substituting the auxiliary expansion into Equation (32) cancels t and gives

$$R(x) = \sum_i p_i x_i + \sum_{i<j} s_{ij} \sigma(x_i - x_j). \tag{33}$$

Substitute Equation (30) into this expression and collect the coefficients of $\Phi(x)$ into w_R. Thus both w_k and w_R are specified by finite sums of the previously defined coefficients. The entries of W_1 and of the rows w_k are integers, because heads and hinges are integer combinations of coordinates. The row w_R is a rational combination of the p_i and s_{ij}, and w_3 consists of the d_k and ± 1.

Order K and set the second hidden layer matrix and output layer vector to

$$W_2 = \begin{pmatrix} (w_k^\top)_{k \in K} \\ w_R^\top \\ -w_R^\top \end{pmatrix}, \qquad w_3 = \begin{pmatrix} (d_k)_{k \in K} \\ 1 \\ -1 \end{pmatrix}, \qquad b_1 = b_2 = b_3 = 0. \tag{34}$$

The second hidden layer outputs $U_k(x)$ and $\sigma(\pm R(x))$. By Equation (8) and Equation (28),

$$w_3^\top \sigma(W_2 \Phi(x)) = \sum_{k \in K} d_k U_k(x) + \sigma(R(x)) - \sigma(-R(x)) = \max_N(x). \tag{35}$$

□

This completes the rational Caterpillar Net. Section 6 bounds q_N by counting its possible normalized descriptions.

3.3 The combinatorial construction of Brunck et al.

The additional building blocks in the direct combinatorial proof of [BHR+26] are the rooted-star functions from Equation (11). In our notation their signed identity is the following.

Theorem 11 (Brunck et al., [BHR+26, Theorem 25])**.** *For $N \geq 3$, there is a linear combination R_N of support functions of polytopes of affine dimension at most $N - 2$ such that*

$$\begin{aligned} \max_N(x) = \frac{(-1)^N}{N!} \sum_{\pi \in \mathrm{Sym}\{0,\ldots,N-1\}} \Bigg[& x_{\pi(0)} + S_{\pi(0),\pi(1)}(x) \\ & + \sum_{\varnothing \neq S \subseteq \{1,\ldots,N-2\}} (-1)^{|S|} \big(x_{\pi(0)} + U_{\pi,S}(x) \big) \Bigg] + R_N(x). \end{aligned} \tag{36}$$

Their construction handles the lower-dimensional remainder by induction. Theorem 19 proves the exact identification of their pyramid summands: nonempty marked sets give translated caterpillar polytopes; the empty marked set gives the translated rooted-star polytope. Translation adds only a linear function, also for every face. Nontrivial faces containing the apex give our caterpillar or rooted-star face functions, up to that linear term. The apex vertex and all base faces have support functions in A_N, since base faces are translated zonotopes generated by coordinate differences. Thus both papers' pyramid families, with all their faces, lie in the joint vocabulary used below. The lower bounds allow arbitrary signed coefficients and arbitrary corrections within that vocabulary; they do not assume either paper's displayed coefficients.

4 Lower bounds

One local test proves both kinds of lower bound: at a tied maximum, take a directional germ and discard lower-dimensional support functions. For expansion size, every tie must be detected by some unit, so we count the ties each unit can detect. For dictionary dimension, we exhibit one locality that detects each path order separately.

Theorem 12 (Exponential count, Proof)**.** *Put $y_0 = \sqrt{3/2} - 1$. Every expansion of $\max_N$, $N \geq 3$, using caterpillar and rooted-star face functions satisfies*

$$|K| \geq (1 + y_0)^N - 1 - N y_0 - \binom{N}{2} y_0^2 = \Omega\big((3/2)^{N/2}\big). \tag{37}$$

Every expansion-size bound in this section permits all face states of its stated building blocks, auxiliary remainders, and arbitrary signed output coefficients. In particular, the bound includes the joint face vocabulary of both constructions in Section 3.3. It does not give a lower bound for unrestricted two-hidden-layer networks.

Theorem 13 (Vocabulary dimension, `Lean_9`, Proof)**.** *For every $N \geq 3$, $q_N = \dim(V_N/A_N) \geq (N-2)!$.*

Thus the whole dictionary has factorial dimension, and $\log q_N = \Theta(N \log N)$ by Equation (97). This concerns the basis used in Section 3.1; an individual expansion of the maximum may have smaller support.

Broader forest dictionaries.

Theorem 14 (Forest expansions, Proof)**.** *Consider an expansion using arbitrary edge states of unit-weight forest functions whose original reduced forests have maximum degree at most $\Delta \geq 2$. All face functions of source-oriented canonical trees may also be included. In particular, the expansion may freely mix these additional functions with caterpillars and the building blocks of Brunck et al., including all their faces. Set $y = (2\Delta + 2)^{-1}$; when $\Delta = 2$, one may instead use $y = y_0 = \sqrt{3/2} - 1$. Then*

$$|K| \geq \sum_{m=3}^{N} \binom{N}{m} y^m = (1 + y)^N - 1 - N y - \binom{N}{2} y^2. \tag{38}$$

The reduced-degree hypothesis is checked before taking faces: face selection cannot increase it. Caterpillars have reduced degree at most two, and rooted stars have reduced degree zero, so neither family is subject to a bound on its original tree degree. The estimates count all terms in the mixture, including both papers' face functions. An ordinary maximum-degree bound also implies the stated reduced-degree bound. The real-head extension and the stronger aggregate estimates are proved mathematically here; the existing Lean count theorem gives a weaker bound.

Arbitrary tree shapes with a source orientation. The source-oriented extension in Theorem 14 includes every $U^{\tau}_{a,s,r,T}$, for every tree shape, source, head root, and face state. At any locality detected by such a face, its retained forest reduces to one smaller source-oriented tree. Equation (110) transfers the same counting bounds without a loss. Thus the family including all these face functions satisfies Equation (37).

Large matchings with arbitrary orientations and weights. Start with any mixture of caterpillars and the building blocks of Brunck et al., including all their faces. More generally, the base family may contain all edge states of reduced-degree-two unit-weight forest functions and all source-oriented canonical tree face functions. Adjoin arbitrary edge states of weighted pseudoforest functions U_{h_j,L_j,c_j}. Suppose each added original graph G_{L_j} has a matching of size at least k, where k is a nonnegative integer with $2k \le N$. This matching hypothesis applies only to the added graphs; caterpillar, rooted-star, and the other base-family face functions remain unrestricted. For $N \ge 9$, every expansion of the maximum in this mixture, with an arbitrary shallow remainder, uses a total number Z of terms satisfying

$$Z \ge \frac{(9/7)^k}{2d_k}, \qquad d_k = 1 + \frac{2k}{7} + \frac{4}{49}\binom{k}{2}. \tag{39}$$

Thus $k = \Omega(N)$ gives an exponential bound. Biases may also be allowed: taking the recession function preserves the homogeneous target and the comparison graphs.

Every orientation and face state of a canonical tree with matching number at least k is included, using the original-graph description following Equation (13). The matching condition concerns that original tree, even if its retained forest has a smaller matching. This includes perfect and near-perfect matchings and stars with every edge subdivided once. More generally, a tree on $n \ge 2$ vertices with L leaves has a matching of size at least $(n - L + 1)/2$, so a fixed positive fraction of nonleaf vertices suffices. Additional cycles with total excess $b = o(N)$ are also permitted when $k = \Omega(N)$; the proof gives the same exponential rate up to $\exp(o(N))$.

Corollary 15 (All tree orientations, `Lean_10`)**.** *For $N \ge 3$, every expansion by canonical tree face functions from Definition 4 has at least $N(N-1)/6$ terms. All tree shapes, edge orientations, and face states are allowed.*

Adding all oriented stars. Canonical oriented star face functions can be added while counting only nonstar terms in the following exponential bounds. The base-family rate remains $\Omega((3/2)^{N/2})$, and Equation (39) holds for $N \ge 15$, counting only nonstar terms. This uses a fixed exact separator and coordinate deletion, described at the end of Appendix C.1. The unrestricted all-tree exponential bound remains open.

5 Why the dictionary spans the maximum

The maximum is the support function of a simplex. To work in its affine dimension, set

$$\Delta := \operatorname{conv}\{0, e_1, \dots, e_{N-1}\} \subseteq H_0 := \{x \in \mathbb{R}^N : x_0 = 0\}. \tag{40}$$

For every $x \in \mathbb{R}^N$,

$$\max_N(x) = x_0 + h_\Delta(x - x_0\mathbf{1}). \tag{41}$$

It suffices to prove that h_Δ lies in the span of the restrictions of $\mathcal{D}$ to H_0: the substitution $x \mapsto x - x_0\mathbf{1}$ fixes every face function, sends coordinate features to $\sigma(\pm(x_i - x_0))$, and preserves the dictionary span by Equation (8).

Henceforth all polytopes lie in H_0, and heads and edge functions are identified with their restricted coefficient vectors. Let $\mathcal{L}_k$ be the real span of support functions of polytopes of affine dimension at most k; thus $\mathcal{L}_0$ consists of linear functions. The *exposed face* of P in direction $u \in H_0$ is

$$P^u := \{p \in P : \langle p, u\rangle = h_P(u)\}. \tag{42}$$

The opposite face P^{-u} consists of the minimizers. Define

$$D_u h_P(x) := h_{P^u}(x) - h_{P^{-u}}(x) \qquad (x \in H_0). \tag{43}$$

Here u selects the faces, while x is the support-function input; $P^0 = P$ gives $D_0 = 0$. For a simplex facet normal, D_u subtracts the opposite vertex's linear support function from that of the facet.

For a finite linear combination g of support functions, define its one-sided directional derivative at u by

$$\begin{aligned} \partial_u g(x) &:= \lim_{t\downarrow 0} \frac{g(u+tx) - g(u)}{t}, \\ \partial_u h_P = h_{P^u}, &\qquad D_u g = \partial_u g - \partial_{-u} g. \end{aligned} \tag{44}$$

This makes D_u independent of the support-function decomposition. When $\langle u, \cdot\rangle$ is constant on P, the exposed faces cancel; otherwise both have smaller dimension. Consequently,

$$D_u \mathcal{L}_k \subseteq \mathcal{L}_{k-1} \qquad (k \geq 1). \tag{45}$$

The signed caterpillar sum. We seek a combination that agrees with h_Δ up to a function in $\mathcal{L}_{N-2}$. Adding $x_{\pi(0)}$ makes the caterpillar function the following maximum

$$x_{\pi(0)} + U_{\pi,S}(x) = \max\left\{ x_{\pi(0)},\, x_{\pi(q+1)} + \sum_{\ell \in L_{\pi,S}} \sigma(\ell(x)) \right\}. \tag{46}$$

Averaging over permutations gives the required symmetry

$$F_N(x) := \frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,r\}} (-1)^{N+|S|} \big(x_{\pi(0)} + U_{\pi,S}(x)\big), \tag{47}$$

The signs cancel the extra exposed-face contributions modulo $\mathcal{L}_{N-3}$, as proved below.

Theorem 16 (Dictionary sufficiency, `Lean_11`, Proof)**.** *For every $N \geq 3$, the dictionary of Definition 7 satisfies*

$$\max_N \in \operatorname{span} \mathcal{D}. \tag{48}$$

Proof of Theorem 16. Define the error by

$$f := F_N|_{H_0} - h_\Delta. \tag{49}$$

Step 16.1 reduces dictionary membership to $f \in \mathcal{L}_{N-2}$; Step 16.2 reduces this to $D_u f \in \mathcal{L}_{N-3}$ for all u. Step 16.3 proves these bounds by cancellation, using induction only when deleting a single exceptional coordinate.

Step 16.1 (Reduce to a lower-dimensional error). Assume temporarily that

$$f = F_N - h_\Delta \in \mathcal{L}_{N-2}. \tag{50}$$

Write $Q_{\pi,S} := Q_{h_{\pi,S},L_{\pi,S}}$. Removing the linear apex terms from Equation (47) gives

$$\frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}h_{Q_{\pi,S}} - h_\Delta \in \mathcal{L}_{N-2}. \tag{51}$$

Put $V := \operatorname{span}\mathcal{D}$. Faces of $Q_{\pi,S}$ already have support functions in V: apex–base faces give Equation (12), base faces give linear functions plus sums of $\sigma(x_i - x_j)$, and the apex alone gives zero. Coordinate features and Equation (8) represent the linear terms and either hinge orientation. Taking another exposed face stays within the same family, so $\partial_u V \subseteq V$ for every u. We use this closure to eliminate the error; induction on the number of simplex vertices supplies the simplex faces that occur along the way.

Claim 16.1 [Simplex faces, `Lean_12`, `Lean_13`, Proof]. For every nonempty $T \subseteq \{0, e_1, \dots, e_{N-1}\}$, the support function of $\Delta_T := \operatorname{conv} T$ lies in the dictionary span.

Start ($|T| = 1$). The support function of a singleton is linear and lies in the dictionary span.

Step ($|T| \geq 2$). Fix T and assume the claim for all smaller nonempty vertex sets. For any current vertex set A with $|A| \geq 2$, remove $v \in A$ using a direction whose value at v exceeds its common value at the other vertices. Writing $\Delta_A := \operatorname{conv} A$, this gives

$$\begin{aligned} u_v &:= \begin{cases} -\sum_{i=1}^{N-1} e_i & v = 0, \\ v & v \neq 0, \end{cases} \\ (\Delta_A)^{u_v} &= \{v\}, \qquad (\Delta_A)^{-u_v} = \Delta_{A\setminus\{v\}}, \\ D_{u_v} h_{\Delta_A}(x) &= \langle v, x\rangle - h_{\Delta_{A\setminus\{v\}}}(x). \end{aligned} \tag{52}$$

Order the vertices outside T as $v_1, \dots, v_{N-|T|}$ and set

$$\varepsilon_T := D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} \left(\frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}h_{Q_{\pi,S}} - h_\Delta \right). \tag{53}$$

Each deletion changes the sign of the remaining simplex term, and subsequent D_u annihilate the linear terms. Hence

$$D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} h_\Delta = (-1)^{N-|T|} h_{\Delta_T} + \ell_T, \tag{54}$$

where ℓ_T is linear. Thus the coefficient of h_{Δ_T} in ε_T is $(-1)^{N-|T|+1}$; all other terms are linear or caterpillar-face support functions and lie in $\operatorname{span}\mathcal{D}$. By Equation (45), $\varepsilon_T \in \mathcal{L}_{|T|-2}$. For the full vertex set the empty composition is the identity.

Claim 16.2 [Finite elimination, Proof]. Let V be a linear subspace of support-function combinations, stable under every ∂_u, and let P be a polytope whose proper-face support functions belong to V. If $g - h_P \in V$ and $g \in \mathcal{L}_k$ with $k < \dim P$, then $h_P \in V$.
To see the mechanism, write $g = \sum_i c_i h_{P_i}$ with $\dim P_i \leq k$. Choose u_i constant on P_i but nonconstant on P. The operator $I - \partial_{u_i}$ kills every face support function of P_i and preserves h_P modulo V. Applying all these operators kills g while preserving its class h_P modulo V. Appendix A.1 gives the details, including why the order of the operators causes no difficulty.

Apply the claim to $P = \Delta_T$ and $g = (-1)^{N-|T|+1}\varepsilon_T$. Its hypotheses follow from the outer induction and $\varepsilon_T \in \mathcal{L}_{|T|-2}$, since $\dim \Delta_T = |T| - 1$. Thus $h_{\Delta_T} \in V$. Taking the full vertex set proves $h_\Delta \in \operatorname{span}\mathcal{D}$ under assumption Equation (50).

Step 16.2 (Why the facet bounds imply the class identity). For $N \geq 4$ and $f = F_N - h_\Delta$, we show that $f \in \mathcal{L}_{N-2}$ follows from

$$D_u f \in \mathcal{L}_{N-3} \qquad \text{for every } u \in H_0. \tag{55}$$

The base zonotope admits the Freudenthal–Kuhn triangulation [Fre42, Kuh60]; coning it to the apex expresses each caterpillar support function as a sum of simplex support functions modulo $\mathcal{L}_{N-2}$. For a simplex, D_u isolates a facet up to the linear support function of its opposite vertex. Thus Equation (55) controls the weighted sums of parallel facets. A common linear reconstruction operator turns these facet cancellations back into a cancellation of the original simplex sum. Define

$$(\mathcal{T}g)(x) := \begin{cases} g(x) & x_1 \geq 0, \\ g(x - x_1 e_1) & x_1 < 0. \end{cases} \tag{56}$$

Claim 16.3 [Simplex reconstruction, `Lean_14`, Proof]. Every $(N-1)$-dimensional simplex $P \subseteq H_0$ satisfies

$$h_P - \mathcal{T}\left(\sum_{F \text{ facet of } P} \operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F \right) \in \mathcal{L}_{N-2}. \tag{57}$$

Here ν_F is an outward normal of F. Its sign against e_1 records whether the facet bounds lines parallel to e_1 from above or below; parallel facets contribute zero. The operator $\mathcal{T}$ fills the segments between a polytope and its projection onto $p_1 = 0$, up to a segment correction for translation; hence $\mathcal{T}\mathcal{L}_k \subseteq \mathcal{L}_{k+1}$. Applying the claim with the same operator to every simplex preserves the cancellations between parallel facets. Their sums lie in $\mathcal{L}_{N-3}$, so their reconstructions lie in $\mathcal{L}_{N-2}$, as required. Appendix A.2 proves the claim and the subdivision and grouping steps.

Claim 16.4 [Class identity, `Lean_15`, Proof]. The class identity Equation (50) holds for every $N \geq 3$.

Step 16.3 (Prove the class identity by induction).

Start ($N = 3$). Here $S = \{1\}$, $q = 1$, and $p(1) = 0$, so Equation (46) gives

$$\begin{aligned} x_{\pi(0)} + U_{\pi,\{1\}}(x) &= \max\bigl\{x_{\pi(0)}, x_{\pi(2)} + \sigma(x_{\pi(1)} - x_{\pi(2)})\bigr\} \\ &= \max\{x_{\pi(0)}, x_{\pi(1)}, x_{\pi(2)}\} = \max_3(x). \end{aligned} \tag{58}$$

All six terms have coefficient $1/6$, so

$$F_3 = \max_3 . \tag{59}$$

On H_0, this gives $f = 0 \in \mathcal{L}_1$.

Step ($N \geq 4$). Assume the class identity for $N - 1$. By Step 16.2, it suffices to prove $D_u f \in \mathcal{L}_{N-3}$ for every $u \in H_0$. The linear apex terms disappear. The exposed simplex faces have the largest and smallest coordinate values u_i, including $u_0 = 0$. In a caterpillar base, $[0, \ell_{ij}]$ remains free when $u_i = u_j$; its maximizing endpoint is ℓ_{ij} when $u_i > u_j$ and 0 when $u_i < u_j$. Minimization reverses these choices.

Which faces can survive? Fix $u \in H_0 \setminus \{0\}$; the direction 0 gives $D_0 f = 0$. A base face is a zonotope, whose support function is linear plus a sum of segment support functions, hence belongs to $\mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$. The apex alone contributes zero. If j tree edges have unequal endpoint values under u, an apex–base face has at most $N-2-j$ free segments and therefore dimension at most $N-1-j$. Thus only $j \le 1$ needs attention.

Write $a = u_{\pi(0)}$. An apex–base face requires the maximum over the base to be zero. For $j = 0$, all tree values equal c, so this condition is $c - a = 0$, forcing $u = 0$. For $j = 1$, deleting the unequal edge leaves constant values $c \neq d$ on its two components. The zero-maximum condition is

$$a = \max\{c, d\} \qquad \text{for a spine edge,} \tag{60}$$

$$a = c + \sigma(c - d) \qquad \text{for a leaf edge, with leaf value } d. \tag{61}$$

The spine case gives two coordinate values. The leaf case does too when $d > c$; when $d < c$, it gives the values $c + t, c - t, c, \ldots, c$, where $t = c - d > 0$. Replacing u by $-u$ treats the opposite face. Consequently, only two-value directions and these symmetric three-value directions can contribute modulo $\mathcal{L}_{N-3}$.

Three values: reflection symmetry. Suppose $u_\alpha = c + t$, $u_\beta = c - t$, and all other coordinates equal c. Let τ exchange α and β. On H_0, its coordinate action is

$$(Rx)_i = x_{\tau(i)} - x_{\tau(0)}, \qquad Rx = x - \lambda(x)u, \qquad \lambda(x) := \frac{x_\alpha - x_\beta}{t}. \tag{62}$$

In particular $Ru = -u$. For $G = F_N|_{H_0}$, permutation symmetry and a common coordinate shift give $G \circ R = G + \ell$ with ℓ linear. Every exposed face in direction v lies in the hyperplane $\langle v, p\rangle = h_P(v)$; hence $\partial_v G(x + sv) = \partial_v G(x) + sG(v)$. Applying this to the derivative of $G \circ R = G + \ell$ at u gives

$$\partial_u G(x) + \ell(x) = \partial_{-u} G(Rx) = \partial_{-u} G(x) + \lambda(x) G(-u). \tag{63}$$

Therefore

$$D_u G = \lambda\, G(-u) - \ell \in \mathcal{L}_0. \tag{64}$$

The two exposed simplex faces are vertices, so $D_u f \in \mathcal{L}_0$ as well.

Handling two distinct values. First suppose that each value occurs at least twice. After the preceding dimension bound, every term still requiring cancellation has a unique split $u_{\pi(1)} = \cdots = u_{\pi(k)} \neq u_{\pi(k+1)} = \cdots = u_{\pi(r+1)}$ with $k \in S$. If the apex label has the latter value, toggle membership of $k-1$ in S. Otherwise, exchange the equal-valued labels $\pi(k+1), \pi(k+2)$ and toggle membership of $k+1$. Repetition of both values gives $k \ge 2$ in the first case and $k + 2 \le r + 1$ in the second, so the operations are defined; the unchanged cut at k keeps the cut set nonempty. The transformed term remains in the same case, and repeating the operation returns the original term. If Φ denotes this involution on the active terms and $w_{\pi,S} := (-1)^{N+|S|}/N!$, then

$$w_{\Phi(\pi,S)} = -w_{\pi,S}, \qquad D_u h_{Q_{\pi,S}} - D_u h_{Q_{\Phi(\pi,S)}} \in \mathcal{L}_1. \tag{65}$$

Inactive fixed points of the global transformation are already in $\mathcal{L}_{N-3}$ by the dimension bound. Summing the active pairs therefore puts $D_u F_N$ in $\mathcal{L}_{N-3}$. Each exposed simplex face has at most $N-2$ vertices and hence dimension at most $N-3$, so $D_u f \in \mathcal{L}_{N-3}$ without using induction.

One exceptional coordinate: the induction hypothesis. Suppose one of the two values occurs only once. Choose τ with $\tau(0)$ the exceptional label. To stay in H_0, relabel and subtract the new zeroth coordinate:

$$B_\tau : H_0 \longrightarrow H_0, \qquad (B_\tau x)_i := x_{\tau(i)} - x_{\tau(0)} \qquad (0 \le i < N). \tag{66}$$

Its inverse is $B_{\tau^{-1}}$, and it sends u to $(0, t, \dots, t)$, $t \neq 0$. Transforming both input and direction preserves every $\mathcal{L}_k$ and changes f only by a linear term, annihilated by D. Rescaling, and reversing the sign of D_u if needed, reduces to $u = (0, 1, \dots, 1)$.

This direction exposes the $(N-2)$-dimensional facet $\Delta' := \mathrm{conv}\{e_1, \dots, e_{N-1}\}$ and the opposite vertex 0. Hence

$$D_u h_\Delta(x) = h_{\Delta'}(x) = \max\{x_1, \dots, x_{N-1}\} \qquad (x \in H_0). \tag{67}$$

The induction hypothesis also applies to unrestricted lower-arity inputs modulo linear functions. Indeed, put $z = (x_1, \dots, x_{N-1})$ and $y = z - x_1\mathbf{1}$. Each summand of F_{N-1} increases by the common shift, so

$$F_{N-1}(z) - \max_{N-1}(z) \equiv F_{N-1}(y) - \max_{N-1}(y) \pmod{\mathcal{L}_0}. \tag{68}$$

The first coordinate of y is zero. Induction and linear precomposition therefore give

$$\big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big) \in \mathcal{L}_{N-3}. \tag{69}$$

Claim 16.5 [Coordinate deletion, `Lean_16`, Proof]. For $N \ge 4$,

$$D_{(0,1,\dots,1)}(F_N|_{H_0}) - \big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1})\big) \in \mathcal{L}_{N-3}. \tag{70}$$

The proof is in Appendix A.3. Its mechanism is that only degree-one tree positions of label 0 survive. Each nonempty lower cut set has N insertion preimages, each contributing $1/N$ of its lower coefficient. The extra empty-cut star sum lies in $\mathcal{L}_{N-3}$ by Boolean cancellation.

Completion of the induction. Combining Equation (67), Equation (69), and Claim 16.5 gives

$$\begin{aligned} D_u f(x) = {} & \big[D_u(F_N|_{H_0})(x) - F_{N-1}(x_1, \dots, x_{N-1})\big] \\ & + \big[F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big]. \end{aligned} \tag{71}$$

Both brackets lie in $\mathcal{L}_{N-3}$, the first by Claim 16.5 and the second by induction. Step 16.2 now gives $f \in \mathcal{L}_{N-2}$, and Step 16.1 gives the dictionary representation. □

6 Network size

The widths in Theorem 1 are worst-case upper bounds: we count all possible descriptions after collecting linear terms into the head and orienting the remaining hinges consistently. The construction retains q_N basis functions; zero padding realizes the stated widths.

The normalized description Equation (26) of a caterpillar function depends only on its undirected spine, apex, and leaf attachments. For m spine vertices these choices give the summand of C_N in Equation (6). Each of the $N-2$ generators has three face states, giving at most $3^{N-2}C_N$ descriptions. The $N(N-1)$ descriptions with no retained hinge and head $x_i - x_j$, $i \neq j$, can be combined with the auxiliary functions. By Lemma 10, the second width is the number of retained descriptions plus two; Appendix B gives the complete count.

Formalization (`Lean_17`). In this working draft, the theorem and equation links point to the local Lean sources. Appendix E prints the network definitions and the main theorem, and describes the verification certificate.

7 CPWL functions

By the Wang–Sun theorem, every CPWL function on $\mathbb{R}^d$ is a signed sum of maxima of $d+1$ affine functions. These affine functions can be taken from any finite family among which the function selects pointwise, without rescaling them.

Theorem (Koutschan–Moser–Ponomarchuk–Schicho, [KMPS23, Theorem 2.4], `Lean_18`, `Lean_19`)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous, and let $g_1, \dots, g_n : \mathbb{R}^d \to \mathbb{R}$ be affine with $f(x) \in \{g_1(x), \dots, g_n(x)\}$ for every x. Then there are $T \in \mathbb{N}$, signs $\varepsilon_t \in \{-1, 1\}$, and maps $\alpha_t : \{0, \dots, d\} \to \{1, \dots, n\}$ with*

$$f(x) = \sum_{t=1}^{T} \varepsilon_t \max_{d+1}\big(g_{\alpha_t(0)}(x), \dots, g_{\alpha_t(d)}(x)\big) \qquad \text{for every } x \in \mathbb{R}^d. \tag{72}$$

The affine functions of every polyhedral description of a CPWL function satisfy the hypothesis. The Lean proof follows Ovchinnikov's max–min form $f = \max_S \min_{j \in S} g_j$, expands it by inclusion–exclusion into an integer combination of maxima over subsets of $\{g_1, \dots, g_n\}$, and reduces each maximum to subsets of at most $d+1$ functions by an affine dependence among $d+2$ gradients.

Reusing the first hidden layer (`Lean_20`, `Lean_21`). Let $d \geq 2$, and let (h_1, h_2) be the widths of Theorem 1 for $N = d+1$. Substituting the arguments of each term of Equation (72) into the Caterpillar Net computes f. All terms use only the features $\sigma(g_a - g_b)$ for $a < b$, $\sigma(\pm g_1)$, and $\sigma(g_a - g_1)$, so one first hidden layer of width $\binom{n}{2} + n + 1$ serves every term. Terms with the same argument set $\alpha_t(\{0, \dots, d\})$ compute the same maximum and are merged, and the auxiliary remainders of all terms form one pair $\sigma(\pm R)$. The second hidden layer has width $s(h_2 - 2) + 2$, where s is the number of distinct argument sets and $s \leq \min\big(T, \sum_{m \leq d+1} \binom{n}{m}\big)$. Thus the second hidden layer holds one copy of the Caterpillar Net units for every distinct maximum; only the first hidden layer and the remainder pair are genuinely shared. The second hidden layer and output layer have rational weights and zero biases, and every first hidden layer neuron applies σ to $\pm(g_a - g_b)$ or $\pm g_a$. Hence rational g_i give rational weights and biases, and linear g_i give zero biases.

8 Conclusion

The Caterpillar Net uses q_N caterpillar face functions with rational coefficients, giving second hidden layer width $q_N + 2$ before padding. Conversely, every such expansion needs $\Omega((3/2)^{N/2})$ caterpillar functions (Theorem 12), so caterpillar building blocks cannot give subexponential second widths. Reducing our upper bound $2^{O(N \log N)}$ to $2^{O(N)}$ remains open. The factorial dimension bound in Theorem 13 limits compression by a basis of the entire dictionary; it does not give a factorial lower bound for representing the single function $\max_N$. All caterpillar face states, including those with disconnected retained forests, are already included in the construction. More generally, a family of $2^{O(N)}$ oriented canonical tree descriptions still has only $2^{O(N)}$ face descriptions: each tree contributes at most 3^{N-2} edge states. A uniform spanning identity for such an exponential-size family remains open. Whether the maximum has two-hidden-layer representations of subexponential width with other second-layer functions remains open.

A Supporting proofs for the dictionary theorem

A.1 Finite elimination of the error

Proof of Claim 16.2. Choose a finite expression $g = \sum_{i=1}^{m} c_i h_{P_i}$ with $\dim P_i \le k < \dim P$. For each i, choose $u_i \in (\operatorname{dir} P_i)^{\perp}$ that is not orthogonal to $\operatorname{dir} P$. Such a direction exists because $\operatorname{dir} P$ is not contained in $\operatorname{dir} P_i$. Set

$$A_i := I - \partial_{u_i}, \qquad \mathcal{A} := A_m \cdots A_1. \tag{73}$$

Every u_i exposes a proper face of P, so $A_i h_P - h_P \in V$. Since $\partial_{u_i} V \subseteq V$, the relation $g - h_P \in V$ therefore implies $\mathcal{A}g - h_P \in V$. On the other hand, each partial composition sends h_{P_i} to a linear combination of support functions of faces of P_i. The direction u_i is constant on every such face F, so $\partial_{u_i} h_F = h_F$ and $A_i h_F = 0$. Thus the ith factor kills all descendants of h_{P_i}, and later factors preserve zero. It follows that $\mathcal{A}g = 0$, hence $h_P \in V$. No commutation of derivatives is required. □

A.2 Reconstruction from opposite exposed faces

We justify Step 16.2: for $N \ge 4$, the bounds Equation (55) imply $f \in \mathcal{L}_{N-2}$. We first subdivide the caterpillar polytopes, then prove Claim 16.3, and finally apply it to the resulting weighted simplex sum.

The base of each $Q_{\pi,S}$ is an affine cube. Order its edge vectors by a bijection $\omega : \{1, \dots, N-2\} \to L_{\pi,S}$, add them one at a time, and adjoin the apex:

$$K_{\pi,S,\omega} := \operatorname{conv}\left(\{0\} \cup \left\{ h_{\pi,S} + \sum_{a=1}^{k} \omega(a) : 0 \le k \le N-2 \right\} \right). \tag{74}$$

These $(N-2)!$ simplices are obtained by coning the *Freudenthal–Kuhn triangulation* of the base to the apex [Fre42, Kuh60]: in cube coordinates, each base simplex occupies one ordering region $1 \ge t_{\omega(1)} \ge \cdots \ge t_{\omega(N-2)} \ge 0$. For a hyperplane cut $P = P_+ \cup P_-$, the support-function identity

$$h_P + h_{P_+ \cap P_-} = h_{P_+} + h_{P_-} \tag{75}$$

shows that subdivision changes the sum of support functions only by support functions of the cutting sections. Hence

$$h_{Q_{\pi,S}} - \sum_{\omega} h_{K_{\pi,S,\omega}} \in \mathcal{L}_{N-2}. \tag{76}$$

Substituting Equation (76) into Equation (47) expresses f as a finite linear combination of support functions of $(N-1)$-dimensional simplices in H_0, plus a remainder in $\mathcal{L}_{N-2}$. The simplex Δ contributes its support function with coefficient -1, and the linear apex terms are included in the remainder.

For a simplex P, an outward facet normal ν_F exposes the facet F on one side and its opposite vertex v_F on the other. Thus

$$D_{\nu_F} h_P(x) = h_F(x) - \langle v_F, x \rangle. \tag{77}$$

Directions exposing no facet on either side give support functions of faces of dimension at most $N-3$. Consequently, Equation (55) says that each weighted, signed sum of parallel-facet support functions belongs to $\mathcal{L}_{N-3}$.

Our assumption controls the weighted facet contributions, whereas our conclusion concerns f. To recover the original sum modulo $\mathcal{L}_{N-2}$, we use the same linear operator for every simplex, so cancellations between

facet contributions survive reconstruction. Use the operator $\mathcal{T}$ defined in Equation (56). The sign in this definition concerns the input x. To interpret it geometrically, first suppose that every point $p \in P$ satisfies $p_1 \geq 0$, and write $\mathrm{pr}(p) := p - p_1 e_1$ for projection onto $p_1 = 0$. Since $\langle p, x\rangle - \langle \mathrm{pr}(p), x\rangle = p_1 x_1$, each point dominates its projection when $x_1 \geq 0$, and its projection dominates when $x_1 < 0$. Consequently,

$$\mathcal{T} h_P(x) = \max\{h_P(x), h_{\mathrm{pr}\, P}(x)\} = h_{\mathrm{conv}(P \cup \mathrm{pr}\, P)}(x). \tag{78}$$

This convex hull fills the segments with endpoints p and its projection, for $p \in P$, so its dimension is at most $\dim P + 1$. For an arbitrary P, choose $c \geq 0$ such that $P + ce_1$ lies in $p_1 \geq 0$. Translation adds cx_1 to the support function, and $\mathcal{T}(x_1) = \sigma(x_1)$, giving

$$\mathcal{T} h_P(x) = h_{\mathrm{conv}((P+ce_1) \cup \mathrm{pr}(P+ce_1))}(x) - c\sigma(x_1). \tag{79}$$

The correction $\sigma(x_1)$ is the support function of the segment $[0, e_1]$. Thus linearity gives

$$\mathcal{T}\mathcal{L}_k \subseteq \mathcal{L}_{k+1}. \tag{80}$$

Proof of Claim 16.3. Let $k = |\{F \text{ facet of } P : \langle \nu_F, e_1\rangle \neq 0\}|$ count facets not parallel to e_1. Boundedness ensures both nonzero signs occur, so $k \geq 2$.

Start ($k = 2$). There is one upper facet F^+ and one lower facet F^-. Every line parallel to e_1 that meets P has its two intersection endpoints on these facets. Thus $h_P(x) = h_{F^+}(x)$ when $x_1 \geq 0$, and $h_P(x) = h_{F^-}(x)$ when $x_1 < 0$. When $x_1 \geq 0$, $\mathcal{T}$ evaluates at x, so subtracting $\mathcal{T}(h_{F^+} - h_{F^-})$ from h_P leaves h_{F^-}. When $x_1 < 0$, $\mathcal{T}$ evaluates both facet support functions at $x - x_1 e_1$, where they agree because the facets have the same projection. Their difference is therefore zero, again leaving h_{F^-}. The reconstruction error is therefore $h_{F^-} \in \mathcal{L}_{N-2}$, proving the claim for $k = 2$.

Step ($k > 2$). Assume the claim for $k - 1$. Choose two facets F_j, F_l with the same nonzero sign against e_1, and let β_j, β_l be their barycentric coordinates: $F_i = \{\beta_i = 0\}$ in P. The constant directional derivatives $b_j := \beta_j(x + e_1) - \beta_j(x)$ and $b_l := \beta_l(x + e_1) - \beta_l(x)$ have the same nonzero sign, opposite to that of the outward normals. The hyperplane $b_l\beta_j - b_j\beta_l = 0$ contains the common ridge and is parallel to e_1. It meets the edge between the two opposite vertices at $\beta_j = b_j/(b_j + b_l) \in (0, 1)$, cutting P into two simplices. Each replaces one chosen facet by a facet parallel to e_1; all other facet normals retain their directions. Both new simplices therefore have $k - 1$ facets not parallel to e_1.

Write $E(Q)$ for the reconstruction error on the left of Equation (57). Let P_+ and P_- be the two new simplices, let $R := P_+ \cap P_-$ be their common cutting facet, and write F_1, F_2 for the chosen facets and $R_F := R \cap F$ for every other facet of P. With $s_F := \mathrm{sgn}\langle \nu_F, e_1\rangle$, applying Equation (75) to P and to every split facet gives the exact residual identity

$$E(P) = E(P_+) + E(P_-) - h_R + \mathcal{T}\left(\sum_{\substack{F \text{ facet of } P \\ F \notin \{F_1, F_2\}}} s_F h_{R_F}\right). \tag{81}$$

Indeed, the inherited facets retain their signs, while the two copies of R are parallel to e_1 and therefore have sign zero. By the induction hypothesis, $E(P_+), E(P_-) \in \mathcal{L}_{N-2}$; moreover, $\dim R = N - 2$ and $\dim R_F \leq N - 3$. Equation Equation (80) therefore puts every term on the right of Equation (81) in $\mathcal{L}_{N-2}$. This proves the claim for k and completes the induction.

□

Conclusion for f. List the simplex terms obtained from Equation (76), including Δ with coefficient -1, as (a_j, P_j). Keeping every occurrence and its original coefficient gives

$$f = \sum_j a_j h_{P_j} + \rho, \qquad \rho \in \mathcal{L}_{N-2}. \tag{82}$$

For each facet F not parallel to e_1, normalize its outward normal by

$$u_F := \frac{\nu_F}{\langle \nu_F, e_1 \rangle}, \qquad \langle u_F, e_1 \rangle = 1. \tag{83}$$

Oppositely oriented parallel normals give the same u_F; the sign of $\langle \nu_F, e_1 \rangle$ is retained in the coefficient. Let $\mathcal{U}$ be the finite set of these normalized normals, and put

$$B_u(P) := \sum_{\substack{F \text{ facet of } P \\ \langle \nu_F, e_1 \rangle \neq 0,\ u_F = u}} \operatorname{sgn}\langle \nu_F, e_1 \rangle h_F, \qquad G_u := \sum_j a_j B_u(P_j). \tag{84}$$

For a simplex P, if u or $-u$ exposes a facet, Equation (77) gives $D_u h_P - B_u(P) \in \mathcal{L}_0$; otherwise both exposed faces have dimension at most $N-3$. Together with $D_u \rho \in \mathcal{L}_{N-3}$ this gives

$$D_u f - G_u = \sum_j a_j \big(D_u h_{P_j} - B_u(P_j) \big) + D_u \rho \in \mathcal{L}_{N-3} \qquad (u \in \mathcal{U}). \tag{85}$$

By Equation (55), each $G_u \in \mathcal{L}_{N-3}$, and therefore $\mathcal{T}(\sum_{u \in \mathcal{U}} G_u) \in \mathcal{L}_{N-2}$ by Equation (80). Summing Equation (57) with the same coefficients shows that f differs from this function by an element of $\mathcal{L}_{N-2}$. Thus $f \in \mathcal{L}_{N-2}$, proving the implication used in Step 16.2.

A.3 Coordinate deletion

Proof of Claim 16.5. We now determine which terms of the signed sum remain for $u = (0, 1, \dots, 1)$. Only label 0 has a different coordinate value from the others. Consequently, a tree-edge generator is nonorthogonal to u exactly when one of its endpoint labels is 0.

If label 0 is the apex label, all tree-edge generators are orthogonal to u. The maximizing face is the whole base $B_{\pi,S}$ and the minimizing face is $\{0\}$, so $D_u h_{Q_{\pi,S}} = h_{B_{\pi,S}} \in \mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$. If label 0 is a tree vertex of degree at least two, at least two segment generators are nonorthogonal to u. The preceding dimension bound then gives $D_u h_{Q_{\pi,S}} \in \mathcal{L}_{N-3}$ as well. We can therefore discard both kinds of term: it suffices to retain descriptions in which label 0 is a tree vertex of degree one.

To express this condition in the permutation notation, let $p := \pi^{-1}(0)$ be the index of label 0 in the full order $(\pi(0), \dots, \pi(N-1))$. When $p \geq 1$, its tree position is $p-1$. A degree-one vertex is either an endpoint of the spine with no attached leaves or a leaf outside the spine. The retained descriptions are thus exactly

$$\begin{aligned} &p = 1, \quad 1 \in S && \text{(root endpoint)}, \\ &2 \leq p \leq N-1, \quad p-1 \notin S && \text{(leaf outside the spine)}, \\ &p = N-1, \quad N-2 \in S && \text{(last spine endpoint)}. \end{aligned} \tag{86}$$

Indeed, the root has only one child precisely when $1 \in S$; a nonroot spine vertex has a parent and also a child unless it occupies the final tree position. Deleting label 0 from the full order now gives an order $\bar{\pi}$ mapping positions $\{0, \dots, N-2\}$ bijectively to labels $\{1, \dots, N-1\}$.

For $0 \le i \le N-3$, let ι_i insert a gap at the prospective pendant position $i+1$. Given a lower cut set $T \subseteq \{1,\dots,N-3\}$, the three active insertion rules are

$$\begin{aligned} \iota_i(t) &:= \begin{cases} t, & t < i+1, \\ t+1, & t \ge i+1, \end{cases} \\ \text{first end:} \quad & (p,S) = (1, \{1\} \cup \{t+1 : t \in T\}), \\ \text{pendant } i\text{:} \quad & (p,S) = (i+2, \{\iota_i(t) : t \in T\}) \qquad (0 \le i \le N-3), \\ \text{last end:} \quad & (p,S) = (N-1, T \cup \{N-2\}). \end{aligned} \tag{87}$$

In every row the upper permutation is uniquely obtained by inserting label 0 at position p in $\bar\pi$. Conversely, with t restricted to $\{1,\dots,N-3\}$, the deletion rules are $T = \{t : t+1 \in S\}$ for the first end, $T = \{t : \iota_i(t) \in S\}$ for pendant i, and $T = S \setminus \{N-2\}$ for the last end. These operations are inverse. The three cases in Equation (86) correspond respectively to the first-end row, the pendant row with $i = p-2$, and the last-end row. Thus a nonempty T has $N-2$ pendant and two endpoint preimages, whereas $T = \varnothing$ has only the two endpoints because upper cut sets are nonempty.

Extend the lower caterpillar notation to $T = \varnothing$ by setting its last spine position and every parent equal to 0, so its head begins at the root $\bar\pi(1)$. For every lower label order $\bar\pi$ put

$$\begin{aligned} C_{\bar\pi,T}(x) &:= x_{\bar\pi(0)} + U_{\bar\pi,T}(x), \\ q_T &:= \max(\{0\} \cup T), \qquad p_T(j) := \max(\{0\} \cup \{t \in T : t < j\}). \end{aligned} \tag{88}$$

For each upper preimage (π, S) of $(\bar\pi, T)$, the insertion identities and the literal factorial coefficients are

$$\begin{aligned} \text{pendant:} \quad & \frac{(-1)^{N+|S|}}{N!} = -\frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \quad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) + C_{\bar\pi,T} \in \mathcal{L}_1, \\ \text{endpoint:} \quad & \frac{(-1)^{N+|S|}}{N!} = \frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \qquad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) - C_{\bar\pi,T} \in \mathcal{L}_1. \end{aligned} \tag{89}$$

Thus every one of the $N-2+2 = N$ preimages contributes $1/N$ of the lower summand modulo $\mathcal{L}_{N-3}$, and together they recover its coefficient in F_{N-1}.

The extra terms are the two endpoint fibres whose deletion leaves $T = \varnothing$, hence a rooted star. By Equation (89), each contributes $1/N$ of the weighted lower summand modulo $\mathcal{L}_1$. For a fixed lower order $\bar\pi$, the nonroot positions are $J = \{0, 2, \dots, N-2\}$, including the original apex position 0. For $a \in J$, let $\bar\pi^{(a)}$ exchange positions 0 and a, keeping the root at position 1; the exchange at $a = 0$ is the identity. Put

$$\begin{aligned} q := |J| = N-2, \qquad & d_a := x_{\bar\pi(1)} - x_{\bar\pi(a)} \quad (a \in J), \\ T_A := \sigma\left(\sum_{a\in A} d_a + \sum_{a \in J\setminus A} \sigma(d_a)\right) \quad & (A \subseteq J), \\ C_{\bar\pi^{(a)},\varnothing} = x_{\bar\pi(a)} + T_{\{a\}} \quad & (a \in J). \end{aligned} \tag{90}$$

The Boolean alternating identity gives

$$\sum_{A\subseteq J} (-1)^{|A|} T_A = 0 \quad \Longrightarrow \quad \sum_{a\in J} T_{\{a\}} = T_\varnothing + \sum_{\substack{A\subseteq J \\ |A|\ge 2}} (-1)^{|A|} T_A. \tag{91}$$

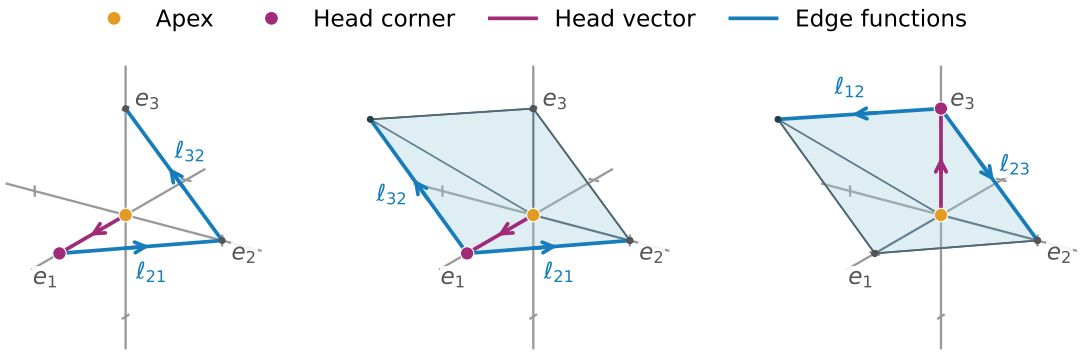
Indeed, if some $d_a \ge 0$, toggling a in A pairs equal terms with opposite signs; if every $d_a < 0$, all terms vanish. The empty term is a sum of support functions of segments. For $|A| \ge 2$, the polytope with support function T_A

has one head direction and $q-|A|$ segment directions, hence dimension at most $1+q-|A| \le q-1 = N-3$. Thus the singleton sum lies in $\mathcal{L}_{N-3}$; the sum of the linear apex coordinates lies in $\mathcal{L}_0$. Each fixed-position exchange is a bijection on lower label orders, so

$$q \sum_{\bar{\pi}} C_{\bar{\pi},\varnothing} = \sum_{\bar{\pi}} \sum_{a \in J} C_{\bar{\pi}^{(a)},\varnothing} \in \mathcal{L}_{q-1}. \tag{92}$$

Since $q > 0$, the full empty-cut sum belongs to $\mathcal{L}_{q-1} = \mathcal{L}_{N-3}$. The empty-cut coefficient $(-1)^{N-1}/(N-1)!$ and the factor $2/N$ from the two endpoint fibres are scalars, so the extra star sum lies there as well. This proves Claim 16.5. □

B Network construction and width bounds



(a) Coordinate differences (b) Original description (c) Normalized description

Figure 3: Coordinate differences and the caterpillar polytope for $\pi = (0, 3, 2, 1)$ and $S = \{1, 2\}$. (a) The head $h = e_1 - e_0$ and edge functions $L = \{\ell_{32} = e_3 - e_2, \ell_{21} = e_2 - e_1\}$ drawn between their labelled endpoints. (b) The same generators translated to the base corner h of $Q_{h,L}$. (c) Normalization (Section 6) gives $h^+ = h + \ell_{32} + \ell_{21} = e_3 - e_0$ and $L^+ = \{\ell_{12} = e_1 - e_2, \ell_{23} = e_2 - e_3\}$, leaving the polytope unchanged. Purple arrows represent the head (the apex–last spine connection before normalization); blue arrows represent the spine edge functions. Orange dots mark the polytope apex and purple dots the chosen head corner. The grey coordinate axes meet at the origin, representing e_0 after restriction to $x_0 = 0$; the positive unit points are labelled e_1, e_2, e_3.

Proof of Theorem 1. Fix $N \ge 3$. The construction in Sections 3.1 and 3.2 already gives rational weights and zero biases. We bound its second hidden layer width $q_N + 2$ by counting normalized descriptions.

Step 1.1 (Count the caterpillar face functions). Use the adjusted heads and normalized descriptions from Section 6. Counting distinct normalized descriptions bounds q_N.

For a caterpillar description (π, S), the spine endpoints are labelled $\pi(1)$ and $\pi(q+1)$, and every label other than $\pi(1)$ and the apex is the child of exactly one edge. Hence the normalized description of $U_{\pi,S}$ is

$$\begin{aligned} h^+_{\pi,S}(x) &= \sum_{\ell_{ij} \in L_{\pi,S}} x_{\max(i,j)} + x_{\pi(1)} + x_{\pi(q+1)} - \sum_{i=0}^{N-1} x_i, \\ L^+_{\pi,S} &= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in L_{\pi,S}\}. \end{aligned} \tag{93}$$

This depends only on the undirected tree, apex, and unordered spine endpoints. For m spine vertices, choose the vertex set in $\binom{N}{m}$ ways, its order up to reversal in $m!/2$ ways, the apex in $N-m$ ways, and each of the $N-m-1$ leaf attachments in m ways. Reversal has no fixed point since $m \geq 2$, so Equation (6) gives

$$\left|\{(h^+_{\pi,S}, L^+_{\pi,S}) : (\pi, S) \text{ a caterpillar description}\}\right| \leq C_N. \tag{94}$$

Reversing an edge swaps its omitted and linear states. Thus choosing states before or after normalization gives the same family $\mathcal{K}_0$ of descriptions, with $|\mathcal{K}_0| \leq 3^{N-2}C_N$. Write $U_k := U_{h,M}$ for $k = (h, M)$. We merge equal descriptions; distinct descriptions may still represent the same function.

Among them are the descriptions $(\ell_{ij}, \varnothing)$ representing $x \mapsto \sigma(x_i - x_j)$, with $i \neq j$: for the all-cut path $S = \{1, \ldots, r\}$ with apex $\pi(0) = j$ and last spine vertex $\pi(r+1) = i$, omitting every edge function gives

$$U^0_{\pi,S} = \sigma \circ \ell_{ij} \qquad (0 \leq i \neq j < N). \tag{95}$$

Let $\mathcal{P} \subseteq \mathcal{K}_0$ consist of descriptions with empty hinge set and head a coordinate difference. The distinct heads ℓ_{ij} give $|\mathcal{P}| \geq N(N-1)$. Put

$$\mathcal{K}_{\text{ess}} := \mathcal{K}_0 \setminus \mathcal{P}. \tag{96}$$

Every face function in $\mathcal{D}$ equals U_k for its normalized description $k \in \mathcal{K}_0$, and by Equation (8) every unit indexed by $\mathcal{P}$ lies in the auxiliary span A_N. Hence the units U_k, $k \in \mathcal{K}_{\text{ess}}$, span V_N modulo A_N. The selected basis descriptions lie in this set, so

$$q_N = |K| \leq |\mathcal{K}_{\text{ess}}| \leq 3^{N-2}C_N - N(N-1). \tag{97}$$

Step 1.2 (Compile the network and establish its widths). The network of Lemma 10 has rational weights, zero biases, first hidden layer width h_1, and second hidden layer width

$$q_N + 2 \leq 3^{N-2}C_N - N(N-1) + 2 = h_2. \tag{98}$$

Zero padding realizes the stated widths exactly.

For the growth rates, count the ordered choices of m spine labels:

$$\binom{N}{m}\frac{m!}{2} = \frac{N!}{2(N-m)!} \leq N^m \qquad (2 \leq m \leq N-1). \tag{99}$$

Using $N - m \leq N$ and $m^{N-m-1} \leq N^{N-m-1}$ in each summand of Equation (6) gives

$$\begin{aligned} C_N &\leq \sum_{m=2}^{N-1} N^m \, N \, N^{N-m-1} = (N-2)N^N \\ &\leq N^{N+1} \leq N^{4N}. \end{aligned} \tag{100}$$

Since $3^{N-2} \leq N^N$ for $N \geq 3$,

$$h_2 \leq 3^{N-2}(N-2)N^N + 2 \leq 2N^{5N}, \tag{101}$$

$$h_1 \leq 4N^2. \tag{102}$$

Taking logarithms gives the asymptotic bounds in Theorem 1. □

C Proofs of the lower bounds

C.1 Proof of the exponential count bound

The shallow functions of Definition 9 form $\mathcal{L}_1$.

Lemma 17 (A common test at a tied maximum)**.** *Fix a finite family of graph functions $U_{h,L,c}$, a set T of $m \geq 3$ labels, and a strict order of the other labels. There is a point p with $p_t = 1$ for $t \in T$ and distinct coordinates below 1 outside T, in the prescribed order, with the following property. Let ι_T extend vectors by zero outside T, fix $t_0 \in T$, and define*

$$\mathcal{J}_{p,T} f := (y \mapsto \partial_p f(\iota_T y))|_{y_{t_0}=0} \pmod{\mathcal{L}_{m-2}}. \tag{103}$$

This linear test annihilates every shallow function and is nonzero on $\max_N$. For a graph function with preactivation A, a nonzero test requires that A vanish identically on this ordered tie stratum. Its tied germ then has the form $\sigma(H(y) + \sum_{\ell_{ij} \in L,\ i,j \in T} c_{\ell_{ij}} \sigma(y_i - y_j))$, with $H(\mathbf{1}_T) = 0$, and the induced comparison graph on T has at most two connected components. An isolated tied label must have nonzero coefficient in H.

Proof. Each preactivation restricts to an affine function of the outside coordinates on the stratum: all comparison signs are fixed. Choose p outside the zero sets of the restrictions that are not identically zero. There are finitely many such proper affine hyperplanes. If $A(p) \neq 0$, the germ of U is zero or shallow. Otherwise the restriction vanishes identically, so every outside coefficient and the constant term vanish. The latter is $H(\mathbf{1}_T)$, since the tied coordinates equal 1.

Directional germs and linear restriction preserve shallow functions. The germ of the maximum is $\max_{i \in T} y_i$; fixing $y_{t_0} = 0$ gives the support function of an $(m-1)$-simplex, which is not in $\mathcal{L}_{m-2}$. Indeed, successive opposite-face differences and facet restrictions, using Equation (45), would otherwise put a segment support function in $\mathcal{L}_0$, although it is not linear.

If the induced graph has c components, its comparison forms span $m - c$ directions. Adjoining H gives at most $m - c + 1$ directions. This bounds the support-function dimension even for signed coefficients: split the preactivation as $h_P - h_Q$ in that span and use $\sigma(h_P - h_Q) = h_{\mathrm{conv}(P \cup Q)} - h_Q$. Thus $c \geq 3$ makes the test zero. If an isolated tied label has zero coefficient in H, all forms annihilate both its coordinate vector and $\mathbf{1}_T$. These independent vectors leave at most $m - 2$ directions, including when that label is the anchor t_0. □

Proof of Theorem 12. We first prove the reduced-degree case of Theorem 14. Taking an edge state cannot increase reduced degree: an original pendant whose edge is retained keeps the same head coefficient and remains a pendant; if its edge is not retained, it is isolated. Thus every new reduced edge was already an old reduced edge. It suffices to bound the resulting forest functions. For a unit-weight forest function $U_{h,L}$, use κ from Definition 3. At an ordered top tie T, let $u(o)$ count the neighbours of o with higher coordinate. Collecting the frozen coefficients gives $\kappa_o - u(o)$ for every $o \notin T$; an isolated tied label has head coefficient κ_t. By Lemma 17, detection requires

$$u(o) = \kappa_o \qquad (o \notin T), \tag{104}$$

$$\kappa_t \neq 0 \qquad (t \in T \text{ with no neighbour in } T). \tag{105}$$

Call a unit *passing* when these conditions hold and its induced comparison graph on T has at most two components. All three conditions are necessary by Lemma 17. Every ordered tie of size at least three is therefore passed by some unit in the expansion.

A tie set is determined by the reduced forest. Let $F = G_L$ and let Φ contain the nonisolated vertices that are not pendants. For a passed tie set T, an original isolate is tied exactly when its κ is nonzero. A pendant is tied exactly when its neighbour is: either mismatch violates Equations (104) and (105). A two-pendant component is entirely tied, since otherwise its lower vertex violates balance. Thus T is determined by $A = T \cap \Phi$, across all cardinalities, and $|T| \geq |A|$.

For fixed T, balance prescribes the out-degrees of $F - T$ when edges point towards the higher endpoint. A forest orientation is uniquely determined by its out-degrees: orient the edge at a leaf, delete the leaf, and repeat. Let $a_U(T)$ be the fraction of outside orders passed by U. For $0 < y \leq 1$, set $F_U(y) = \sum_{|T|\geq 3} a_U(T) y^{|T|}$. Averaging the covering inequality over outside orders and summing over all tied sizes gives

$$\sum_{m=3}^{N} \binom{N}{m} y^m \leq \sum_{U} F_U(y). \tag{106}$$

It remains to bound F_U by products on disjoint blocks of Φ. The injection $T \mapsto A$ and $y^{|T|} \leq y^{|A|}$ allow each subset of each block to be counted once. Relative orders in disjoint blocks are independent.

Edges and three-vertex paths. When the reduced degree is at most two, partition each nontrivial path into edges and, if its size is odd, one three-vertex path. An isolated vertex of Φ has a pendant neighbour; fix one. These fixed neighbours are distinct and outside Φ. If the singleton is outside T, its fixed edge has a prescribed order, giving the factor $1/2 + y$. For an edge and a three-vertex path, respectively, the factors are

$$E(y) = \tfrac{1}{2} + 2y + y^2, \qquad P(y) = \tfrac{1}{3} + 2y + 3y^2 + y^3. \tag{107}$$

For the empty subset of a three-vertex path, a prescribed orientation allows at most two of the six orders. Selecting only the middle vertex leaves no comparison, whereas selecting only a leaf leaves one prescribed edge; their combined contribution is $2y$. All other coefficients count subsets without further restrictions. At $y_0 = \sqrt{3/2} - 1$, we have $E(y_0) = 1$, $P(y_0) = 5/6 + y_0/2 < 1$, and $1/2 + y_0 < 1$. Thus $F_U(y_0) \leq 1$, and Equation (106) proves Equation (37) and the degree-two case of Equation (38).

General reduced degree. An inclusion-minimal edge cover of the nonisolated reduced vertices is a disjoint union of stars: it contains no three-edge path, whose middle edge could be removed. Each star has at most Δ leaves. Whenever its centre and a leaf are both outside A, retain one edge; its prescribed order costs a factor $1/2$. A star with s leaves therefore has factor

$$W_s(y) = (\tfrac{1}{2} + y)(1 + y)^s + \tfrac{1}{2} y^s. \tag{108}$$

Use $y = (2\Delta + 2)^{-1} \leq 1/6$. Singletons and one-leaf stars have factors at most $2/3$ and $31/36$. For $2 \leq s \leq \Delta$, $W_s(y) \leq (1/2 + y)e^{1/2-y} + y^2/2 \leq (2/3)e^{1/3} + 1/72 < 1$. Hence again $F_U(y) \leq 1$, proving Equation (38). Removing pendants leaves at most the spine of a caterpillar, and no edge of a rooted star. The face-stability argument therefore proves Equation (37) for both families and all their faces. □

All faces of source-oriented trees. We first bound a full source-oriented function, then transfer the bound to every face. The same path blocks work whenever a covering subforest on Φ has maximum degree two and each isolated vertex either has a pendant neighbour or is forced into every passed tie set. A forced singleton has factor y. For $U_{a,s,r,T}$, only the source s and head root r can be *out-leaves*, meaning leaves with $\kappa = 1$. Root the tree at r. Include every out-leaf edge, then choose one child edge at each nonleaf having neither a pendant nor an out-leaf neighbour. These edges lie in the reduced forest. Each vertex has at most one selected child edge: only s can be an out-leaf child, an out-leaf root has just one child, and a vertex choosing a child

had no selected child edge. It also has at most one parent edge. The covering forest therefore has maximum degree two. Its isolated vertices in Φ either have a pendant neighbour or are leaves with $\kappa \notin \{0, 1\}$, which balance forces into T. Thus $F_U(y_0) \le 1$ for full source-oriented functions. At $y = 1/2$, the same blocks give $F_U(1/2) \le (\sqrt{7}/2)^N$, as checked in the matching argument below.

Now take an arbitrary face $U^{\tau}_{a,s,r,T} = U_{h',M}$, and let F be its retained forest on the nonapex labels, including isolates. For each such vertex put $e_v = \kappa_v - \deg_F(v) + 1$. Initially $e = \mathbf{1}_s + \mathbf{1}_r$. Fixing an edge $i \to j$ linearly adds one to e_i; omitting it adds one to e_j. Thus the e_v are nonnegative integers, and every component D of F satisfies

$$\sum_{v \in D} e_v = 1 + \sum_{v \in D} h'_v. \tag{109}$$

This follows by summing $\kappa_v = h'_v + \operatorname{outdeg}_F(v)$ and using $|E(D)| = |D| - 1$.

At a passed tie B, the apex must be tied, since its coefficient is -1 and it has no comparison edge. The component condition forces $B \setminus \{a\}$ into one retained component C. Every other component is wholly untied; summing its balance equations gives $\sum_{v \in D} h'_v = 0$. Since the nonapex head coefficients sum to 1, the component C has head sum 1. By Equation (109), it has total charge two and every other component has charge one. This identifies C independently of the tie. If there is no such unique component, or if C is a singleton, the face passes no tie of size at least three.

Otherwise $e|_C = \mathbf{1}_u + \mathbf{1}_v$, allowing $u = v$. Reverse the comparisons in C to point away from u, adjusting the head by Equation (8). This preserves κ, and the new head coefficient at $w \in C$ is $\kappa_w - (\deg_C(w) - 1 + \mathbf{1}_{w=u}) = \mathbf{1}_{w=v}$. Together with the apex this is a canonical source-oriented function U_C on the smaller label set $C \cup \{a\}$. Every passed outside order restricts to a passed order for U_C. A uniform outside order induces a uniform relative order on these labels, so no interleaving factor is lost:

$$\begin{aligned} F_{U^\tau}(y) &\le F_{U_C}(y), \qquad F_{U^\tau}(y_0) \le 1, \\ F_{U^\tau}(1/2) &\le (\sqrt{7}/2)^{|C|+1} \le (\sqrt{7}/2)^N. \end{aligned} \tag{110}$$

For $y = (2\Delta + 2)^{-1} \le y_0$, monotonicity also gives $F_{U^\tau}(y) \le 1$. This proves the source-oriented face extension of Theorem 14 and supplies its capacity in the mixed bound.

Large matchings in the same counting argument. First remove biases, if present, by taking the recession function $\lim_{t\to\infty} F(tx)/t$. This preserves the target, comparison graphs, and number of terms. For an added weighted pseudoforest function with graph $G = G_L$, now let $a_U(T)$ be the indicator that $G[T]$ has at most two components. Together with the base-family passing fractions, these capacities still satisfy Equation (106), by the common test.

Choose T by selecting each label independently with probability p. Orient a pseudoforest with in-degree at most one, independently of its ReLU orientations. For each of k disjoint matching edges, the event that its parent is absent and its child present gives a vertex with in-degree zero in $G[T]$. Each induced component has at most one such vertex. These independent events each have probability $p(1-p)$, so

$$\Pr\{c(G[T]) \le 2\} \le \rho^k \left(1 + ka + \binom{k}{2} a^2\right), \qquad \rho = 1 - p(1-p), \quad a = \frac{p(1-p)}{\rho}. \tag{111}$$

Set $p = 1/3$, equivalently $y = p/(1-p) = 1/2$, so $\rho = 7/9$ and $a = 2/7$. The added units cover probability at most $\rho^k d_k$. For a weighted edge state, absorb its linear terms into the real head and retain nonretained comparisons in the original graph with coefficient zero, as in Equation (13). Thus the matching hypothesis is unchanged. For the base units put $\tau = \sqrt{7}/2$. The singleton factors at $y = 1/2$ are at most τ, while $E(1/2) = 7/4 = \tau^2$ and $P(1/2) = 53/24 \le \tau^3$. Hence $F_U(1/2) \le \tau^{|\Phi|} \le \tau^N$ for reduced-degree-two

units and full source-oriented units. Equation (110) gives the same bound for all source-oriented faces. A base unit covers probability at most $(2/3)^N \tau^N = \rho^{N/2} \le \rho^k$. Multiplying Equation (106) by $(2/3)^N$ consequently yields

$$\Pr\{\mathrm{Bin}(N, 1/3) \ge 3\} \le Z \rho^k d_k. \tag{112}$$

The left side is increasing in N and exceeds $1/2$ at $N = 9$. This proves Equation (39).

For completeness, define the excess of a graph by $b(G) = \sum_C \max(0, |E(C)| - |V(C)|)$ over its connected components. Extend a maximum matching to a spanning forest and add one omitted edge in each originally cyclic component. The resulting pseudoforest preserves the matching and omits exactly $b(G)$ edges. Restoring those edges reduces the induced component count by at most $b(G)$. For excess at most b, replace d_k by $\sum_{j=0}^{\min(k,b+2)} \binom{k}{j}(2/7)^j$. When $k = \Omega(N)$ and $b = o(N)$, this sum is $\exp(o(N))$. The leaf criterion follows from a minimum vertex cover C of a tree, whose size equals its maximum matching number ν. The nonleaf vertices outside C form an independent set I. Their at least $2|I|$ incident edges lie in the forest on $I \cup C$, so $2|I| \le |I| + \nu - 1$ and $n - L \le 2\nu - 1$.

All tree orientations. For Corollary 15, apply the common test to triples. Balance forces the apex into the tie, since its coefficient is -1 and no comparison edge meets it. The apex is isolated, so the other two tied labels must be the endpoints of a retained edge. Each unit has at most $N - 2$ such edges. Covering all triples gives $\binom{N}{3} \le (N - 2)|K|$, proving the claim.

Adding every oriented star. Work in full coordinates. For a label j, let ι_j insert a zero at that coordinate and put $\mathcal{T}_j g = (\partial_{-e_j} g - \partial_{e_j} g) \circ \iota_j$. This linear operator sends $\mathcal{L}_d$ into $\mathcal{L}_{d-1}$ for $d \ge 1$ and annihilates $\mathcal{L}_0$: if coordinate j is constant on a polytope, its two exposed faces coincide; otherwise both have smaller dimension. Projection under restriction cannot increase dimension. Moreover, $\mathcal{T}_j \max_m = \max_{m-1}$, since the simplex faces are its facet opposite e_j and the vertex e_j.

For a canonical star, deletion of a nonroot leaf gives the smaller star modulo $\mathcal{L}_1$, with sign $+$ for $\sigma(x_l - x_c)$ and sign $-$ for $\sigma(x_c - x_l)$. If the deleted leaf is the head root, the latter orientation transfers the head to the centre with sign $+$; the former gives zero modulo $\mathcal{L}_1$. Deleting the apex leaves a zonotope, and deleting the centre fixes every segment, so these cases also give zero modulo $\mathcal{L}_1$. Thus successive deletions preserve the span of canonical stars modulo the lower-dimensional terms.

At five labels, a six-flag certificate gives a linear functional that annihilates every canonical oriented star and $\mathcal{L}_3$, and takes value 120 on the simplex. The independent vertex checker checks all 640 labelled stars, with every head root and orientation, by successive maxima and minima over explicit vertex sets. Composing this functional with $m - 5$ coordinate deletions therefore gives such a separator at every $m \ge 5$. A nonhinge state already has dimension at most $m - 2$, so all star face states are covered too.

Assign every term whose function belongs to the star family to the added star class, even if it was originally listed in another class. At any tie of size at least five, a surviving top-dimensional star germ is again a canonical star: each untied component has a unique attachment to the tied subtree, and summing its zero frozen coefficients transports the head to that attachment if it contains the original root. Apply the separator to the local expansion. The star terms vanish, so some nonstar term must cover the locality. Thus Equation (106) holds with the sum starting at $m = 5$ and only nonstar terms on the right. At y_0 this gives the same exponential rate, subtracting only a polynomial of degree four. For the matching mixture, replace the tail in Equation (112) by $\Pr\{\mathrm{Bin}(N, 1/3) \ge 5\}$, which exceeds $1/2$ for $N \ge 15$. This proves the claims without counting the added stars.

C.2 Proof of the dimension bound

Proof of Theorem 13. Put $r = N - 2$ and $b = N - 1$. For every permutation π fixing 0 and b, take the path $\pi(1), \dots, \pi(r), b$, with apex 0 and every edge a hinge:

$$U_\pi(x) = \sigma\Big(x_b - x_0 + \sum_{j=1}^{r} \sigma(x_{\pi(j)} - x_{\pi(j+1)})\Big) \in V_N. \tag{113}$$

For an order ρ, put $c = \rho(r)$ and choose

$$p^\rho_0 = p^\rho_b = 1, \qquad p^\rho_{\rho(j)} = 1 + j - r \quad (1 \le j \le r). \tag{114}$$

The three labels $0, b, c$ tie at the top. Use the test $\mathcal{J}_\rho = \mathcal{J}_{p^\rho,\{0,b,c\}}$ from Equation (103), anchored at 0. It annihilates A_N and indeed every shallow function. For $\pi = \rho$, only the last edge can activate in the tied germ, which is $\sigma(y_b - y_0 + \sigma(y_c - y_b)) = \max(y_0, y_b, y_c) - y_0$. Its anchored class is the nonzero triangle class $v = [\max(0, s, t)]$ modulo $\mathcal{L}_1$. For $\pi \neq \rho$, the path order has a strict descent in ρ-height, so its outer preactivation is strictly positive at p^ρ. Its germ is therefore shallow. Identifying the two free tied coordinates with (s, t) gives

$$\mathcal{J}_\rho U_\pi = \begin{cases} v, & \pi = \rho, \\ 0, & \pi \neq \rho. \end{cases} \tag{115}$$

Applying $\mathcal{J}_\rho$ to any relation $\sum_\pi c_\pi U_\pi \in A_N$ gives $c_\rho v = 0$, hence $c_\rho = 0$. The $r!$ paths are independent modulo A_N, proving $q_N \ge (N-2)!$. □

D Relation to the construction of Brunck et al.

The combinatorial proof of [BHR+26, Part 2] cones zonotopes $B_{v,\pi,M}$ to pyramids and expresses the simplex support function as a signed sum of their support functions modulo lower-dimensional terms [BHR+26, Lemma 22]. This appendix identifies these building blocks. For a nonempty set M of marked positions, the pyramid has the support function of a translated caterpillar polytope, with apex v, label order π, and cut set $M - 1$. For $M = \varnothing$ it gives a rooted star, which is not a caterpillar function. We use the indexing of [BHR+26, Section 7.1], with their n written as N and $[N]$ identified with our label set $\{0, \dots, N-1\}$. Positions of an ordering are numbered from 1.

Definition 18 (Zonotopes of Brunck et al., `Lean_22`, `Lean_23`)**.** Let $N \ge 3$, let v be a label, let $\pi = (\pi_1, \dots, \pi_{N-1})$ be an ordering of the remaining labels, and let $M \subseteq \{2, \dots, N-1\}$. For $p \in \mathbb{R}^N$ put $u_i(p) := \sum_{1 \le j < i} p_{\pi_j}$. Following [BHR+26, Equation (3)], define

$$H_{-v} := \Big\{p \in \mathbb{R}^N : p_v = 0,\ \sum_k p_k = 1\Big\}, \tag{116}$$

$$B_{v,\pi,M} := \left\{p \in H_{-v} : \begin{array}{ll} 0 \le u_i(p) \le 1 & \text{for } i \in M, \\ -1 \le p_{\pi_i} \le 0 & \text{for } i \in \{2, \dots, N-1\} \setminus M \end{array}\right\}. \tag{117}$$

The pyramid with apex e_v is $\mathrm{pyr}_{e_v}(B_{v,\pi,M}) := \mathrm{conv}(\{e_v\} \cup B_{v,\pi,M})$, with support function $h_{v,\pi,M}$ as in Equation (19). Its last marked position and the marked parent of a position i are

$$q := \max(\{1\} \cup M), \qquad r(i) := \max\big(\{1\} \cup \{j \in M : j < i\}\big) \qquad (2 \le i \le N-1). \tag{118}$$

For $M \neq \varnothing$, $q = \max M$. Rooted stars are as in Equation (11).

Theorem 19 (Relation to Brunck et al., `Lean_24`, `Lean_25`, `Lean_26`, `Lean_27`, `Lean_28`, `Lean_29`, `Lean_30`, `Lean_31`, Proof)**.** *Let v, π, and M be as in Definition 18, and let q and $r(i)$ be as in Equation* (118)*.*

(a) The base is the zonotope

$$B_{v,\pi,M} = e_{\pi_q} + \sum_{i=2}^{N-1} \big[0, e_{\pi_{r(i)}} - e_{\pi_i}\big]. \tag{119}$$

(b) For every $x \in \mathbb{R}^N$,

$$h_{v,\pi,M}(x) = x_v + \sigma\Big(x_{\pi_q} - x_v + \sum_{i=2}^{N-1} \sigma\big(x_{\pi_{r(i)}} - x_{\pi_i}\big)\Big). \tag{120}$$

(c) Let $M \neq \varnothing$. Define the caterpillar description (π', S) by $\pi'(0) := v$, $\pi'(j) := \pi_j$ for $1 \le j \le N-1$, and $S := \{i-1 : i \in M\}$. Then $|S| = |M|$ and

$$h_{v,\pi,M}(x) = x_v + U_{\pi',S}(x) \qquad (x \in \mathbb{R}^N). \tag{121}$$

Hence $h_{v,\pi,M} - x_v = U_k$ for some $k \in \mathcal{K}_0$. Conversely, every caterpillar description (π', S) arises in this way from $v = \pi'(0)$, $\pi_j = \pi'(j)$, and $M = \{j+1 : j \in S\} \neq \varnothing$.

(d) Let $M = \varnothing$. Then $h_{v,\pi,\varnothing}(x) = x_v + S_{v,\pi_1}(x)$. No rooted star is a caterpillar function: $S_{a,c} \neq U_k$ for all labels $a \neq c$ and all $k \in \mathcal{K}_0$. In particular, $h_{v,\pi,\varnothing} - x_v \neq U_k$ for every $k \in \mathcal{K}_0$.

By Equation (20), the right-hand side of Equation (121) is the support function of the translated caterpillar polytope $e_v + Q_{h_{\pi',S},L_{\pi',S}}$. The family of [BHR+26] therefore consists of the caterpillar polytopes, translated by e_v, and of the pyramids over rooted stars. The sign $(-1)^{|M|}$ of [BHR+26, Lemma 22] equals $(-1)^{|S|}$. The theorem compares building blocks only. It makes no statement about corrections or network coefficients.

Proof of Theorem 19. Write $I := \{2, \dots, N-1\}$ and $g_i := e_{\pi_{r(i)}} - e_{\pi_i}$ for $i \in I$. Every element of $\{1\} \cup M$ lies in $\{1, \dots, N-1\}$, so q and all $r(i)$ are positions. Since $M \subseteq I$, the definitions give

- $1 \le r(i) < i$ for $i \in I$, and $i \le r(l)$ whenever $i \in M$ and $i < l$,
- $i \le q$ for $i \in M$,
- $q \neq i$ and $r(l) \neq i$ for $i \in I \setminus M$ and all l.

Step 19.1 (Coordinates of the zonotope). For $t \in [0,1]^I$ let $z(t) := e_{\pi_q} + \sum_{l \in I} t_l g_l$. Every e_{π_j} has coordinate sum 1 and vanishes at v, so $z(t) \in H_{-v}$. For $i \in I \setminus M$ the coordinate $z(t)_{\pi_i}$ collects the term $-t_i$ and no other term, since π_i is neither π_q nor any $\pi_{r(l)}$. Hence $z(t)_{\pi_i} = -t_i$. For $i \in M$, the prefix sum u_i counts a vector e_{π_j} exactly when $j < i$. Thus e_{π_q} contributes 0, because $q \ge i$. A generator g_l with $l < i$ contributes $1 - 1 = 0$, because $r(l) < l < i$. A generator with $l > i$ contributes $0 - 0$, because $r(l) \ge i$. Finally g_i contributes t_i, because $r(i) < i$. Hence $u_i(z(t)) = t_i$. Therefore $z(t) \in B_{v,\pi,M}$.

Step 19.2 (The coordinates determine the point). Let $p, p' \in H_{-v}$ with $u_i(p) = u_i(p')$ for $i \in M$ and $p_{\pi_i} = p'_{\pi_i}$ for $i \in I \setminus M$. Put $d := p - p'$ and $U_i := \sum_{1 \le j < i} d_{\pi_j}$. Then $U_1 = 0$, and $U_N = \sum_k d_k - d_v = 0$. Descending from $i = N-1$ to $i = 2$, either $i \in M$ and $U_i = 0$ by assumption, or $i \notin M$ and $U_i = U_{i+1} - d_{\pi_i} = U_{i+1} = 0$. Hence $d_{\pi_j} = U_{j+1} - U_j = 0$ for $1 \le j \le N-1$, and $d_v = 0$. Thus $p = p'$. This is the invertibility in [BHR+26, Claim 21].

Step 19.3 (Proof of (a)). Step 19.1 shows that the right-hand side of Equation (119) lies in $B_{v,\pi,M}$. Conversely, let $p \in B_{v,\pi,M}$, and set $t_i := u_i(p)$ for $i \in M$ and $t_i := -p_{\pi_i}$ for $i \in I \setminus M$. The inequalities in Equation (117) say $t \in [0,1]^I$. By Step 19.1, the points p and $z(t)$ of H_{-v} have the same coordinates, so $p = z(t)$ by Step 19.2.

Step 19.4 (Proof of (b)). A linear function attains the same supremum on a set and on its convex hull. Hence $h_{v,\pi,M}(x) = \max(x_v, \sup_{p \in B_{v,\pi,M}} \langle p, x \rangle)$. Put $b_l := x_{\pi_{r(l)}} - x_{\pi_l}$. By (a), $\langle z(t), x \rangle = x_{\pi_q} + \sum_l t_l b_l$, and $t_l b_l \le \sigma(b_l)$ for $0 \le t_l \le 1$, with equality for $t_l = \mathbb{1}_{\{b_l \ge 0\}}$. So the supremum over $B_{v,\pi,M}$ equals $A := x_{\pi_q} + \sum_l \sigma(b_l)$. Then Equation (7) gives $\max(x_v, A) = x_v + \sigma(A - x_v)$, which is Equation (120).

Step 19.5 (Proof of (c)). With π' and S as in (c), position j of the caterpillar carries the label $\pi'(j+1) = \pi_{j+1}$, so caterpillar position j is the position $j+1$ of [BHR+26]. The map $i \mapsto i - 1$ is a bijection from M onto S, hence $|S| = |M|$, and it carries $\{1\} \cup M$ onto $\{0\} \cup S$ and $\{1\} \cup \{j \in M : j < i\}$ onto $\{0\} \cup \{j \in S : j < i - 1\}$. Comparing with Equation (15), the last spine position is $q - 1$ and the parent of caterpillar position $i - 1$ is $r(i) - 1$. By Equations (16) and (17), the head is $x_{\pi_q} - x_v$ and the edge functions are $x_{\pi_{r(i)}} - x_{\pi_i}$, $i \in I$. So the right-hand side of Equation (120) is $x_v + U_{\pi',S}(x)$. If $M \neq \varnothing$, then $S \neq \varnothing$ and $0 \notin S$, so (π', S) is a caterpillar description, and its normalized description $k \in \mathcal{K}_0$ satisfies $U_k = U_{\pi',S}$ by Equation (8). The converse inverts these bijections.

Step 19.6 (Proof of (d)). For $M = \varnothing$, $q = 1$ and $r \equiv 1$, and $\{\pi_i : i \in I\}$ is the set of labels other than v and π_1. So Equation (120) becomes $x_v + S_{v,\pi_1}(x)$. Let $a \neq c$ and evaluate at $x = e_c$. Every inner term of $S_{a,c}(e_c)$ equals $\sigma(1)$, so $S_{a,c}(e_c) = 1 + (N-2) = N - 1$. Let $k \in \mathcal{K}_0$. Then U_k is the normalization of a face function $U^{\tau}_{\pi',S}$, and $U_k = U^{\tau}_{\pi',S}$ by Equation (8). Since $\ell \le \sigma \circ \ell$, $0 \le \sigma \circ \ell$, and σ is monotone, $U^{\tau}_{\pi',S} \le U_{\pi',S}$. At $x = e_c$, every head and edge function takes values in $\{-1, 0, 1\}$. If $c = \pi'(0)$, the argument of the outer σ in $U_{\pi',S}(e_c)$ is -1, so $U_{\pi',S}(e_c) = 0$. Otherwise c is the label of a caterpillar position m. The head contributes at most 1, and every inner hinge at most 1. If $m \ge 1$, the inner hinge of the edge into position m is $\sigma(x_{\pi'(p(m)+1)} - 1) = 0$, so the value is at most $1 + (N-3)$. If $m = 0$, the head contributes 0, because $q \ge \min S > 0$, so the value is at most $N - 2$. Thus $U_k(e_c) \le N - 2 < S_{a,c}(e_c)$. □

E Lean definitions and main theorem

The real parameter arrays in Equation (4) have dimensions

$$\begin{aligned} W_1 &\in \mathbb{R}^{h_1 \times N}, \quad b_1 \in \mathbb{R}^{h_1}, \\ W_2 &\in \mathbb{R}^{h_2 \times h_1}, \quad b_2 \in \mathbb{R}^{h_2}, \\ w_3 &\in \mathbb{R}^{h_2}, \qquad b_3 \in \mathbb{R}, \end{aligned} \tag{122}$$

Sources (`Definitions.lean`, `Theorems.lean`). The listings specify the network, widths, and correctness for every real input. They reproduce declarations from these files at revision `8b4d74ab66f4`, omitting imports, documentation comments, and the theorem's proof body; these remain in the linked sources.

E.1 Network and width definitions

```
namespace MaxCaterpillar

open BigOperators

abbrev Point (N : ℕ) := Fin N → ℝ
```

```
noncomputable def relu (t : ℝ) : ℝ := max 0 t

structure Network (N h₁ h₂ : ℕ) where
  W₁ : Fin h₁ → Fin N → ℝ
  b₁ : Fin h₁ → ℝ
  W₂ : Fin h₂ → Fin h₁ → ℝ
  b₂ : Fin h₂ → ℝ
  w₃ : Fin h₂ → ℝ
  b₃ : ℝ

noncomputable def Network.eval {N h₁ h₂ : ℕ} (net : Network N h₁ h₂)
    (x : Point N) : ℝ :=
  (∑ k, net.w₃ k * relu
    ((∑ j, net.W₂ k j * relu
      ((∑ i, net.W₁ j i * x i) + net.b₁ j)) + net.b₂ k))
    + net.b₃

def Network.HasRationalWeights {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  (∀ j i, ∃ q : ℚ, (q : ℝ) = net.W₁ j i) ∧
    (∀ k j, ∃ q : ℚ, (q : ℝ) = net.W₂ k j) ∧
    ∀ k, ∃ q : ℚ, (q : ℝ) = net.w₃ k

def Network.HasZeroBiases {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  net.b₁ = 0 ∧ net.b₂ = 0 ∧ net.b₃ = 0

noncomputable def Maximum (N : ℕ) [NeZero N] (x : Point N) : ℝ :=
  Finset.univ.sup'
    ⟨⟨0, Nat.pos_of_ne_zero (NeZero.ne N)⟩,
      Finset.mem_univ _⟩ x

def firstWidth (N : ℕ) : ℕ := N.choose 2 + N + 1

def coneCount (N : ℕ) : ℕ :=
  ∑ m in Finset.Icc 2 (N - 1),
    N.choose m * (m.factorial / 2) * (N - m)
      * m ^ (N - m - 1)

def secondWidth (N : ℕ) : ℕ :=
  coneCount N * 3 ^ (N - 2) - N * (N - 1) + 2

end MaxCaterpillar
```

E.2 Caterpillar functions and expansions

These definitions, also in `Definitions.lean`, state the results of Section 4. A description $k = (h, M)$ is stored in reduced coordinates, and `realUnitY` evaluates U_k on $\mathbb{R}^N$.

```
abbrev FaceKey (r : ℕ) :=
  MaxNN.Caterpillar.Corrections.RealKey (r + 1)

noncomputable def caterpillarDescriptions (r : ℕ) :
    Finset (FaceKey r) :=
  MaxNN.Caterpillar.Corrections.normalizedKeys r

noncomputable def caterpillarFunction {r : ℕ}
    (k : FaceKey r) : Point (r + 1 + 1) → ℝ :=
  MaxNN.Caterpillar.Corrections.realUnitY k

noncomputable def auxiliaryFunction (N : ℕ)
    (j : Fin (MaxNN.Width.W N)) : Point N → ℝ :=
  fun x => MaxNN.Width.feat N x j

def IsCaterpillarExpansion (r : ℕ)
    (K : Finset (FaceKey r)) : Prop :=
  K ⊆ caterpillarDescriptions r ∧
    ∃ (d : FaceKey r → ℝ)
      (a : Fin (MaxNN.Width.W (r + 1 + 1)) → ℝ),
      ∀ x, Maximum (r + 1 + 1) x =
        ∑ k in K, d k * caterpillarFunction k x +
          ∑ j, a j * auxiliaryFunction (r + 1 + 1) j x

noncomputable def auxiliarySpan (N : ℕ) :
    Submodule ℝ (Point N → ℝ) :=
  Submodule.span ℝ (Set.range (auxiliaryFunction N))

noncomputable def caterpillarSpan (r : ℕ) :
    Submodule ℝ (Point (r + 1 + 1) → ℝ) :=
  Submodule.span ℝ
    ((caterpillarFunction '' ↑(caterpillarDescriptions r))
      ∪ Set.range (auxiliaryFunction (r + 1 + 1)))

noncomputable def caterpillarDim (r : ℕ) : ℕ :=
  FiniteDimensional.finrank ℝ (↑(caterpillarSpan r) /
    (auxiliarySpan (r + 1 + 1)).comap
      (caterpillarSpan r).subtype)
```

E.3 The main theorems

The assumption `[NeZero N]` expresses positive arity. For $N \geq 3$ the first statement is Theorem 1; the natural-number width definitions also cover $N = 1, 2$. With $N = r + 2$, the next three are Theorem 12, and the last is Theorem 13. All are in `Theorems.lean`.

```
theorem caterpillar_net (N : ℕ) [NeZero N] :
    ∃ net : Network N (firstWidth N) (secondWidth N),
      (∀ x, net.eval x = Maximum N x) ∧
        net.HasRationalWeights ∧ net.HasZeroBiases

theorem caterpillar_expansion_choose_le (r : ℕ)
    {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K)
    {k : ℕ} (hk : 3 ≤ k) (hkN : k ≤ r + 1 + 1) :
    (r + 1 + 1).choose k ≤ 6 ^ k * K.card

theorem caterpillar_expansion_exponential (r : ℕ)
    (hr : 1 ≤ r) {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K) :
    7 ^ (r + 1 + 1) ≤
      (r + 1 + 1 + 1) * 6 ^ (r + 1 + 1) * K.card

theorem caterpillar_expansion_quadratic (r : ℕ)
    (hr : 1 ≤ r) {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K) :
    (r + 1 + 1) * (r + 1) ≤ 6 * K.card

theorem caterpillarDim_ge (r : ℕ) (hr : 1 ≤ r) :
    r.factorial ≤ caterpillarDim r
```

E.4 The building blocks of Brunck et al.

The definitions of Definition 18, in the indexing of [BHR+26, Section 7.1], are in `Competing/Defs.lean`. An ordering is a bijection from the $N-1$ positions onto the labels other than v, and `Ordering.pos` returns π_i at position i, $1 \le i \le N-1$.

```
noncomputable def supportFunction {n : ℕ}
    (P : Set (Fin n → ℝ)) (y : Fin n → ℝ) : ℝ :=
  sSup ((fun p => ∑ k, p k * y k) '' P)

def pyramid {n : ℕ} (q : Fin n → ℝ)
    (P : Set (Fin n → ℝ)) : Set (Fin n → ℝ) :=
  convexHull ℝ (insert q P)

def zonotope {n : ℕ} (a : Fin n → ℝ) (I : Finset ℕ)
    (g : ℕ → Fin n → ℝ) : Set (Fin n → ℝ) :=
  {p | ∃ t : ℕ → ℝ, (∀ i ∈ I, 0 ≤ t i ∧ t i ≤ 1) ∧
    p = a + ∑ i in I, t i • g i}

abbrev BHRSY.Ordering (n : ℕ) (v : Fin n) :=
  Fin (n - 1) ≃ {k : Fin n // k ≠ v}

def BHRSY.Ordering.pos {v : Fin n} (π : Ordering n v)
```

```
    (i : ℕ) : Fin n :=
  if h : i - 1 < n - 1 then (π ⟨i - 1, h⟩).1 else v

def BHRSY.hyperplane (v : Fin n) : Set (Fin n → ℝ) :=
  {p | p v = 0 ∧ ∑ k, p k = 1}

noncomputable def BHRSY.prefixSum {v : Fin n}
    (π : Ordering n v) (p : Fin n → ℝ) (i : ℕ) : ℝ :=
  ∑ j in Finset.Ico 1 i, p (π.pos j)

def BHRSY.parallelepiped (v : Fin n) (π : Ordering n v)
    (M : Finset ℕ) : Set (Fin n → ℝ) :=
  {p | p ∈ hyperplane v ∧
    (∀ i ∈ M, 0 ≤ prefixSum π p i ∧ prefixSum π p i ≤ 1) ∧
    ∀ i ∈ Finset.Icc 2 (n - 1), i ∉ M →
      -1 ≤ p (π.pos i) ∧ p (π.pos i) ≤ 0}

def BHRSY.paramid (v : Fin n) (π : Ordering n v)
    (M : Finset ℕ) : Set (Fin n → ℝ) :=
  pyramid (Pi.single v 1) (parallelepiped v π M)

noncomputable def BHRSY.paramidSupport (v : Fin n)
    (π : Ordering n v) (M : Finset ℕ) : (Fin n → ℝ) → ℝ :=
  supportFunction (paramid v π M)

def BHRSY.lastMarked (M : Finset ℕ) : ℕ :=
  (insert 1 M).max' (Finset.insert_nonempty _ _)

def BHRSY.markedParent (M : Finset ℕ) (i : ℕ) : ℕ :=
  (insert 1 (M.filter fun j => j < i)).max'
    (Finset.insert_nonempty _ _)
```

The dictionary uses the word $\pi' = (v, \pi_1, \ldots, \pi_{N-1})$ (`Ordering.word`), the cut set $S = M - 1$ (`cutSet`), and the description $(h_{\pi',S}, L_{\pi',S})$ of a word (`caterpillarWordDescription`). The rooted star is `BHRSY.rootedStar`. With $N = r + 2$, `Theorems.lean` states Theorem 19 as follows.

```
theorem bhrsy_parallelepiped_eq_zonotope {r : ℕ}
    {v : Fin (r + 1 + 1)} (π : BHRSY.Ordering (r + 1 + 1) v)
    {M : Finset ℕ} (hM : M ⊆ Finset.Icc 2 (r + 1)) :
    BHRSY.parallelepiped v π M =
      zonotope (Pi.single (π.pos (BHRSY.lastMarked M)) 1)
        (Finset.Icc 2 (r + 1))
        (fun i => Pi.single (π.pos (BHRSY.markedParent M i)) 1
          - Pi.single (π.pos i) 1)

theorem bhrsy_support_eq {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) {M : Finset ℕ}
    (hM : M ⊆ Finset.Icc 2 (r + 1)) (y : Point (r + 1 + 1)) :
    BHRSY.paramidSupport v π M y =
```

```
      y v + MaxCaterpillar.relu
        (y (π.pos (BHRSY.lastMarked M)) - y v +
          ∑ i in Finset.Icc 2 (r + 1), MaxCaterpillar.relu
            (y (π.pos (BHRSY.markedParent M i)) - y (π.pos i)))

theorem bhrsy_support_eq_word {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) {M : Finset ℕ}
    (hM : M ⊆ Finset.Icc 2 (r + 1)) :
    (BHRSY.cutSet (r := r) M).card = M.card ∧
      ∀ y : Point (r + 1 + 1), BHRSY.paramidSupport v π M y =
        y v + caterpillarFunction
          (caterpillarWordDescription π.word (BHRSY.cutSet M)) y

theorem bhrsy_support_mem_caterpillar {r : ℕ}
    {v : Fin (r + 1 + 1)} (π : BHRSY.Ordering (r + 1 + 1) v)
    {M : Finset ℕ} (hM : M ⊆ Finset.Icc 2 (r + 1))
    (hne : M.Nonempty) :
    ∃ k ∈ caterpillarDescriptions r, ∀ y,
      BHRSY.paramidSupport v π M y = y v + caterpillarFunction k y

theorem bhrsy_support_empty {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) (y : Point (r + 1 + 1)) :
    BHRSY.paramidSupport v π ∅ y =
      y v + BHRSY.rootedStar v (π.pos 1) y

theorem rootedStar_ne_caterpillarFunction {r : ℕ}
    {a c : Fin (r + 1 + 1)} (hac : a ≠ c) {k : FaceKey r}
    (hk : k ∈ caterpillarDescriptions r) :
    BHRSY.rootedStar a c ≠ caterpillarFunction k
```

The same file contains the converse `bhrsy_word_mem`, the special cases `bhrsy_parallelepiped_eq_zonotope_of_nonempty` and `bhrsy_parallelepiped_empty`, the statement `bhrsy_support_empty_not_caterpillar`, and the dichotomy `bhrsy_family`.

E.5 Proof routes in the formalization

The paper selects the coefficients of its single Caterpillar Net by the finite rational evaluation system in Section 3.1. The public Lean theorem `caterpillar_net` certifies rational coefficient existence and the stated width using an internal $\mathbb{Q}$-linear projection argument. The evaluation-matrix proof itself is not a new formalized declaration.

The formalized count proof uses star blocks and gives the weaker bound $(7/6)^N/(N+1)$. The written proof in Appendix C.1 uses one generic germ and sums over all tie sizes, sharpening the estimate and allowing real heads with unit inner coefficients. The formalized vocabulary-dimension proof uses a third mixed difference; Appendix C.2 gives a shorter proof using the same germ as the count argument.

The finite elimination in Claim 16.2 also replaces induction on residual dimension in the formalization. The relative face lemma proves membership in the relative face span without an additional linear remainder; the previous statement follows by inclusion.

The keyword `noncomputable` allows classical definitions of real quantities. The theorem is an exact representation statement; it does not by itself assert an efficient algorithm for extracting numerical weights.

E.6 Verification scope

The repository's historical verification certificate, dated 24 October 2026, records a build with Lean 4.4.0 and checks 122 public statements against their exact types. Its permitted axioms are `propext`, `Classical.choice`, and `Quot.sound`. The checked interface includes dictionary sufficiency, rational expansions, network correctness with rational weights and zero biases, the width bounds, the compilation and compression theorems, the Wang–Sun and Koutschan–Moser–Ponomarchuk–Schicho representations, and the network theorems for sums of maxima and CPWL functions with their rationality and bias statements. The equation and theorem links identify supporting declarations for the mathematical arguments in the paper. The certificate and its source hashes are available in the linked verification record. The subsequent consolidation of the Lean modules is not covered by that certificate: the current sources require an updated exact-statement manifest and a fresh verification run. In particular, the rational evaluation system, the common local test, the sharpened count bounds, and the source-orientation and face-closure, matching-number, and arbitrary-star extensions are mathematical arguments in this paper, not claims checked by the historical certificate. A separate exact-statement audit checks the simplified lower-bound declarations, dictionary sufficiency, the class identity, and finite face elimination with the same permitted axioms. The mathematical arguments are provided in the main text and Appendices A to C; the formalization is an additional certificate of the formal statements.

AI methodology statement

Claude Code and Codex were used for mathematical exploration and proof search, development of the Lean formalization, and drafting and revision of the manuscript. Their use materially affected the construction and proofs in Sections 3 to 7 and Appendices A to E. The author set the research objectives, directed the searches, and revised the mathematical presentation. The accompanying Lean certificate checks the formal statements and their axiom dependencies; it does not certify the prose exposition, literature comparison, or claims of originality. The author is responsible for the contents of the paper.

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